

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	V
Subject Code & Name	ICS1512 & Machine Learning Algorithms Laboratory		
Academic year	2025-2026 (Odd)	Batch:2023-2028	Due date:

Experiment 3: Email Spam or Ham Classification using Naive Bayes, KNN, and SVM

1 Aim:

To develop and evaluate supervised machine learning models using K-Nearest Neighbors (KNN) and Naive Bayes (Bernoulli, Multinomial, Gaussian) classifiers. To analyze and compare the performance of these models on a given dataset using accuracy and other evaluation metrics.

2 Libraries used:

- Numpy
- Pandas
- Matplotlib
- Scikit-learn
- Seaborn

3 Objective:

- To implement KNN and Naive Bayes algorithms for classification, including preprocessing, training, and testing using appropriate techniques such as cross-validation.
- To assess the models using metrics like accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC, and to present a comparative analysis.

4 Naive Bayes Code:

naive

August 5, 2025

0.0.1 Importing Required Libraries

```
[19]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler # or MinMaxScaler, if used
from sklearn.preprocessing import LabelEncoder # if categorical labels
from sklearn.naive_bayes import BernoulliNB, MultinomialNB, GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, \
    classification_report
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.model_selection import cross_val_score, KFold
```

0.0.2 Loading dataset

```
[2]: df = pd.read_csv('spambase.csv')
```

0.0.3 Basic info

```
[3]: print("Dataset Info:\n", df.info())
print("\nFirst 5 rows:\n", df.head())
print("\nMissing Values:\n", df.isnull().sum())
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 4601 entries, 0 to 4600

Data columns (total 58 columns):

#	Column	Non-Null Count	Dtype
0	word_freq_make	4601 non-null	float64
1	word_freq_address	4601 non-null	float64
2	word_freq_all	4601 non-null	float64
3	word_freq_3d	4601 non-null	float64
4	word_freq_our	4601 non-null	float64
5	word_freq_over	4601 non-null	float64
6	word_freq_remove	4601 non-null	float64

7	word_freq_internet	4601	non-null	float64
8	word_freq_order	4601	non-null	float64
9	word_freq_mail	4601	non-null	float64
10	word_freq_receive	4601	non-null	float64
11	word_freq_will	4601	non-null	float64
12	word_freq_people	4601	non-null	float64
13	word_freq_report	4601	non-null	float64
14	word_freq_addresses	4601	non-null	float64
15	word_freq_free	4601	non-null	float64
16	word_freq_business	4601	non-null	float64
17	word_freq_email	4601	non-null	float64
18	word_freq_you	4601	non-null	float64
19	word_freq_credit	4601	non-null	float64
20	word_freq_your	4601	non-null	float64
21	word_freq_font	4601	non-null	float64
22	word_freq_000	4601	non-null	float64
23	word_freq_money	4601	non-null	float64
24	word_freq_hp	4601	non-null	float64
25	word_freq_hpl	4601	non-null	float64
26	word_freq_george	4601	non-null	float64
27	word_freq_650	4601	non-null	float64
28	word_freq_lab	4601	non-null	float64
29	word_freq_labs	4601	non-null	float64
30	word_freq_telnet	4601	non-null	float64
31	word_freq_857	4601	non-null	float64
32	word_freq_data	4601	non-null	float64
33	word_freq_415	4601	non-null	float64
34	word_freq_85	4601	non-null	float64
35	word_freq_technology	4601	non-null	float64
36	word_freq_1999	4601	non-null	float64
37	word_freq_parts	4601	non-null	float64
38	word_freq_pm	4601	non-null	float64
39	word_freq_direct	4601	non-null	float64
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42	word_freq_original	4601	non-null	float64
43	word_freq_project	4601	non-null	float64
44	word_freq_re	4601	non-null	float64
45	word_freq_edu	4601	non-null	float64
46	word_freq_table	4601	non-null	float64
47	word_freq_conference	4601	non-null	float64
48	char_freq_%3B	4601	non-null	float64
49	char_freq_%28	4601	non-null	float64
50	char_freq_%5B	4601	non-null	float64
51	char_freq_%21	4601	non-null	float64
52	char_freq_%24	4601	non-null	float64
53	char_freq_%23	4601	non-null	float64
54	capital_run_length_average	4601	non-null	float64

```

55 capital_run_length_longest 4601 non-null int64
56 capital_run_length_total    4601 non-null int64
57 class                       4601 non-null int64

```

```
dtypes: float64(55), int64(3)
```

```
memory usage: 2.0 MB
```

```
Dataset Info:
```

```
None
```

```
First 5 rows:
```

```

      word_freq_make word_freq_address word_freq_all word_freq_3d \
0          0.00          0.64          0.64          0.0
1          0.21          0.28          0.50          0.0
2          0.06          0.00          0.71          0.0
3          0.00          0.00          0.00          0.0
4          0.00          0.00          0.00          0.0

```

```

      word_freq_our word_freq_over word_freq_remove word_freq_internet \
0          0.32          0.00          0.00          0.00
1          0.14          0.28          0.21          0.07
2          1.23          0.19          0.19          0.12
3          0.63          0.00          0.31          0.63
4          0.63          0.00          0.31          0.63

```

```

      word_freq_order word_freq_mail ... char_freq_%3B char_freq_%28 \
0          0.00          0.00 ...          0.00          0.000
1          0.00          0.94 ...          0.00          0.132
2          0.64          0.25 ...          0.01          0.143
3          0.31          0.63 ...          0.00          0.137
4          0.31          0.63 ...          0.00          0.135

```

```

      char_freq_%5B char_freq_%21 char_freq_%24 char_freq_%23 \
0          0.0          0.778          0.000          0.000
1          0.0          0.372          0.180          0.048
2          0.0          0.276          0.184          0.010
3          0.0          0.137          0.000          0.000
4          0.0          0.135          0.000          0.000

```

```

      capital_run_length_average capital_run_length_longest \
0          3.756          61
1          5.114          101
2          9.821          485
3          3.537          40
4          3.537          40

```

```

      capital_run_length_total class
0          278          1
1         1028          1
2         2259          1

```

3	191	1
4	191	1

[5 rows x 58 columns]

Missing Values:

word_freq_make	0
word_freq_address	0
word_freq_all	0
word_freq_3d	0
word_freq_our	0
word_freq_over	0
word_freq_remove	0
word_freq_internet	0
word_freq_order	0
word_freq_mail	0
word_freq_receive	0
word_freq_will	0
word_freq_people	0
word_freq_report	0
word_freq_addresses	0
word_freq_free	0
word_freq_business	0
word_freq_email	0
word_freq_you	0
word_freq_credit	0
word_freq_your	0
word_freq_font	0
word_freq_000	0
word_freq_money	0
word_freq_hp	0
word_freq_hpl	0
word_freq_george	0
word_freq_650	0
word_freq_lab	0
word_freq_labs	0
word_freq_telnet	0
word_freq_857	0
word_freq_data	0
word_freq_415	0
word_freq_85	0
word_freq_technology	0
word_freq_1999	0
word_freq_parts	0
word_freq_pm	0
word_freq_direct	0
word_freq_cs	0
word_freq_meeting	0

```

word_freq_original      0
word_freq_project       0
word_freq_re            0
word_freq_edu           0
word_freq_table         0
word_freq_conference    0
char_freq_%3B           0
char_freq_%28           0
char_freq_%5B           0
char_freq_%21           0
char_freq_%24           0
char_freq_%23           0
capital_run_length_average 0
capital_run_length_longest 0
capital_run_length_total  0
class                   0
dtype: int64

```

0.0.4 Handling missing values

```

[6]: imputer = SimpleImputer(strategy='mean')
     df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)

```

0.0.5 Splitting of feature and target

```

[7]: X = df_imputed.drop('class', axis=1)
     y = df_imputed['class']

```

0.0.6 Checking Distribution

```

[8]: X = df.drop('class', axis=1)  # Assuming 'class' is the target

# Choose a few features to visualize
sample_features = X.columns[:5]  # First 5 features

# Plot histograms with Gaussian curve overlay
plt.figure(figsize=(15, 10))
for i, feature in enumerate(sample_features):
    plt.subplot(3, 2, i + 1)
    data = X[feature]

    # Plot histogram
    count, bins, ignored = plt.hist(data, bins=30, density=True, alpha=0.6,
    color='skyblue', edgecolor='black')

    # Plot normal distribution curve
    '''mu, std = data.mean(), data.std()

```

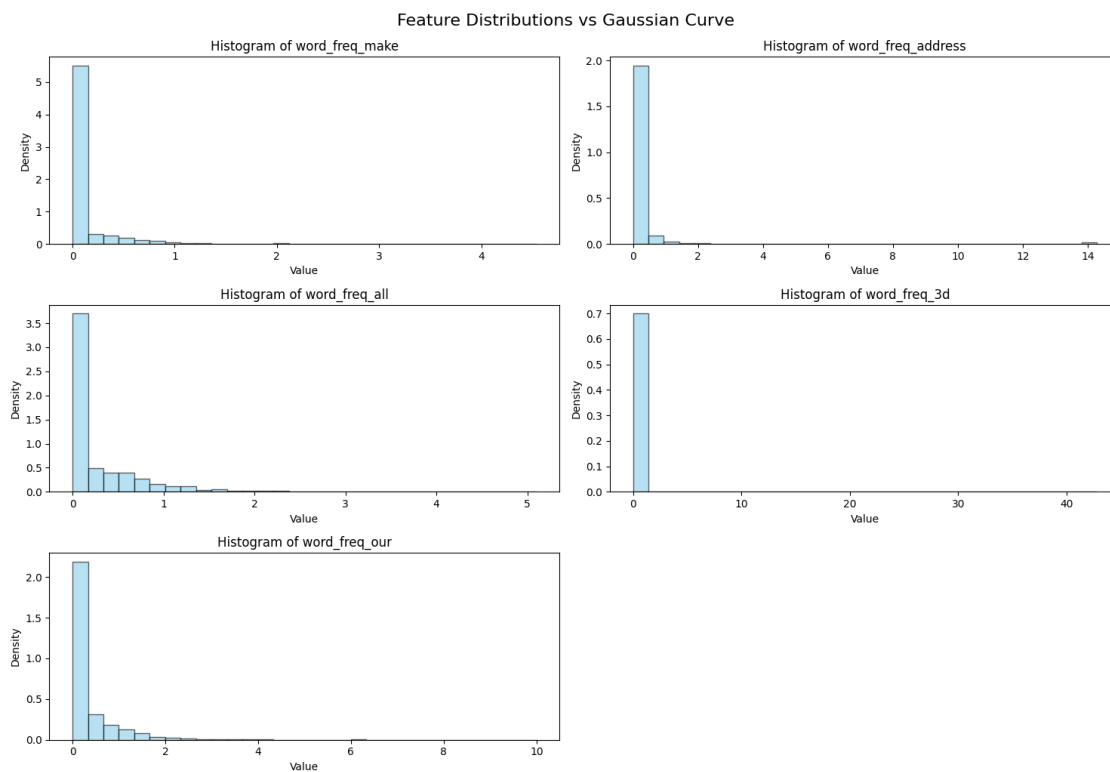
```

xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = 1 / (std * np.sqrt(2 * np.pi)) * np.exp(-(x - mu)**2 / (2 * std**2))
plt.plot(x, p, 'r', linewidth=2)'''

plt.title(f'Histogram of {feature}')
plt.xlabel('Value')
plt.ylabel('Density')

plt.tight_layout()
plt.suptitle('Feature Distributions vs Gaussian Curve', fontsize=16, y=1.02)
plt.show()

```



0.0.7 Applying min max scaling

```

[10]: scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

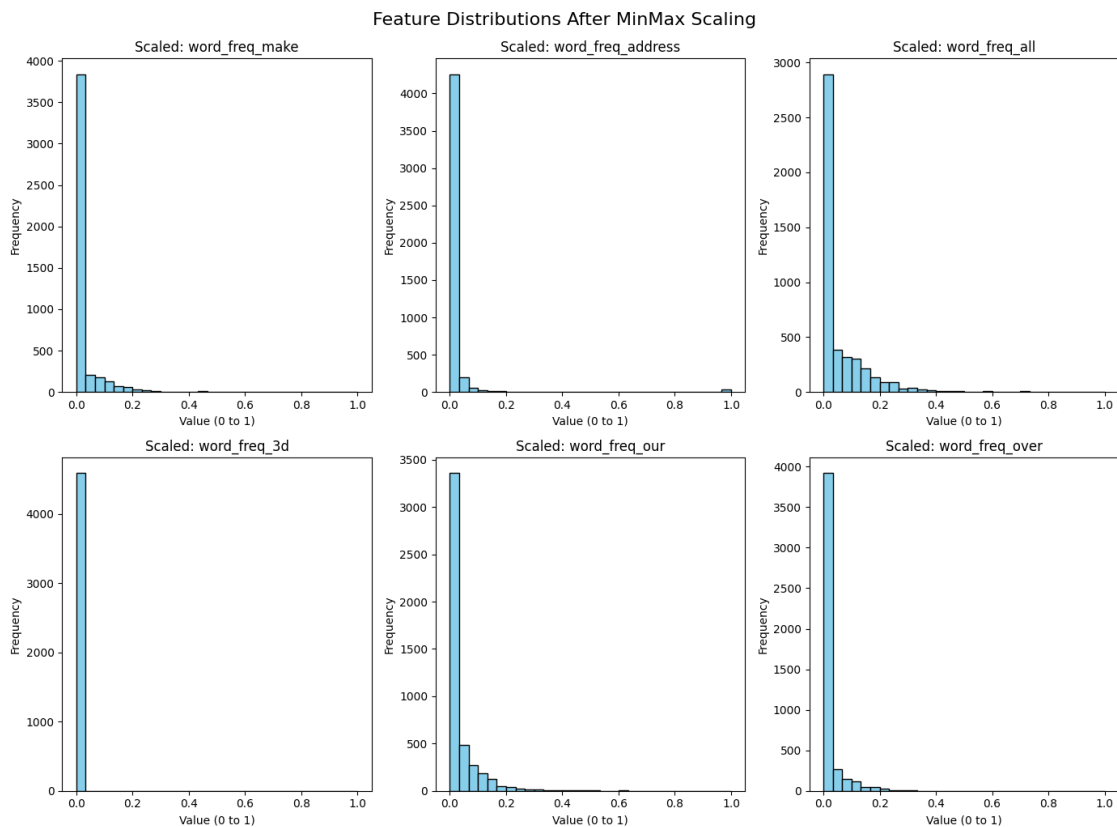
```

0.1 Plots

0.1.1 Histogram

```
[11]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))
for i in range(6): # first 6 features as an example
    plt.subplot(2, 3, i + 1)
    plt.hist(X_scaled[:, i], bins=30, color='skyblue', edgecolor='black')
    plt.title(f'Scaled: {X.columns[i]}')
    plt.xlabel('Value (0 to 1)')
    plt.ylabel('Frequency')
plt.tight_layout()
plt.suptitle('Feature Distributions After MinMax Scaling', fontsize=16, y=1.02)
plt.show()
```

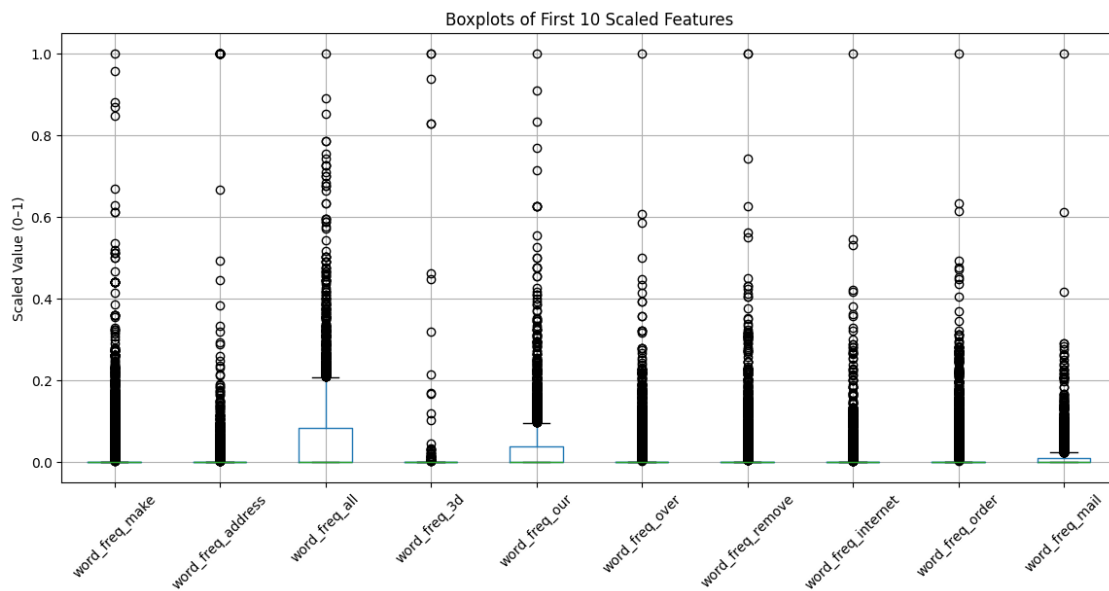


0.1.2 Boxplot

```
[12]: import pandas as pd

# Convert scaled array back to DataFrame for easier plotting
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)

plt.figure(figsize=(14, 6))
X_scaled_df.iloc[:, :10].boxplot(rot=45)
plt.title('Boxplots of First 10 Scaled Features')
plt.ylabel('Scaled Value (0-1)')
plt.grid(True)
plt.show()
```



0.1.3 Correlation HeatMap

```
[13]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Assuming X_scaled_df is your scaled DataFrame
corr_matrix = X_scaled_df.corr()

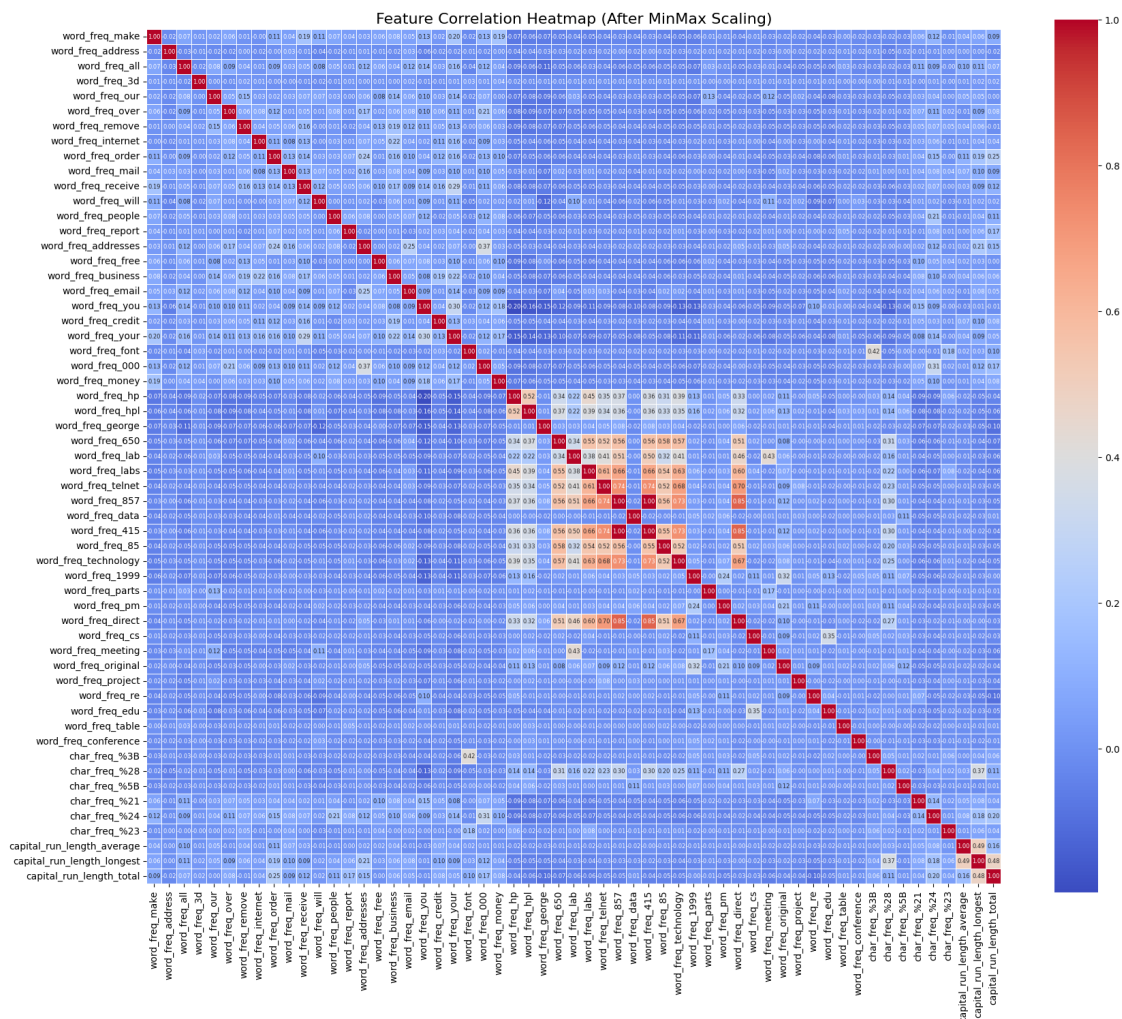
plt.figure(figsize=(18, 15)) # Bigger figure
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
```

```

square=True,
annot=True,                                     # Show values inside squares
fmt=".2f",                                       # Format to 2 decimal places
linewidths=0.5,                                # Thin grid lines
annot_kws={"size": 6}                           # Smaller font size
)

plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.title("Feature Correlation Heatmap (After MinMax Scaling)", fontsize=16)
plt.tight_layout()
plt.show()

```



0.1.4 Model Training

```
[14]: from sklearn.metrics import (
        accuracy_score, precision_score, recall_score,
        f1_score, confusion_matrix, roc_curve, auc,
        classification_report
    )

    # Split the data again (or reuse your earlier split)
    from sklearn.model_selection import train_test_split
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled, y, test_size=0.3, stratify=y, random_state=42
    )

[16]: def evaluate_model(name, model, X_test, y_test, results):
        y_pred = model.predict(X_test)

        # For ROC and AUC, use predict_proba
        try:
            y_proba = model.predict_proba(X_test)[:, 1]
        except:
            y_proba = y_pred # fallback if proba not available

        acc = accuracy_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred, zero_division=0)
        rec = recall_score(y_test, y_pred, zero_division=0)
        f1 = f1_score(y_test, y_pred, zero_division=0)

        # Print Confusion Matrix and Classification Report
        print(f"\n=== {name} ===")
        print("Confusion Matrix:")
        print(confusion_matrix(y_test, y_pred))
        print("\nClassification Report:")
        print(classification_report(y_test, y_pred, zero_division=0))

        # ROC and AUC
        fpr, tpr, _ = roc_curve(y_test, y_proba)
        roc_auc = auc(fpr, tpr)

        plt.plot(fpr, tpr, label=f'{name} (AUC = {roc_auc:.2f})')

        # Append results
        results.append({
            'Model': name,
            'Accuracy': acc,
            'Precision': prec,
            'Recall': rec,
```

```

        'F1 Score': f1,
        'AUC': roc_auc
    })

```

```

[17]: # Store results
results = []

# --- BernoulliNB ---
bnb = BernoulliNB()
bnb.fit(X_train, y_train)
evaluate_model("BernoulliNB", bnb, X_test, y_test, results)

# --- MultinomialNB ---
mnb = MultinomialNB()
mnb.fit(X_train, y_train)
evaluate_model("MultinomialNB", mnb, X_test, y_test, results)

# --- GaussianNB ---
gnb = GaussianNB()
gnb.fit(X_train, y_train)
evaluate_model("GaussianNB", gnb, X_test, y_test, results)

# === ROC Curve Plot ===
plt.title('ROC Curves')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

# === Final Comparison Table ===
results_df = pd.DataFrame(results)
print("\n=== Comparison Table ===")
print(results_df)

```

=== BernoulliNB ===

Confusion Matrix:

```

[[778  59]
 [ 93 451]]

```

Classification Report:

	precision	recall	f1-score	support
0.0	0.89	0.93	0.91	837
1.0	0.88	0.83	0.86	544
accuracy			0.89	1381

macro avg	0.89	0.88	0.88	1381
weighted avg	0.89	0.89	0.89	1381

=== MultinomialNB ===

Confusion Matrix:

```
[[811  26]
 [119 425]]
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.87	0.97	0.92	837
1.0	0.94	0.78	0.85	544
accuracy			0.90	1381
macro avg	0.91	0.88	0.89	1381
weighted avg	0.90	0.90	0.89	1381

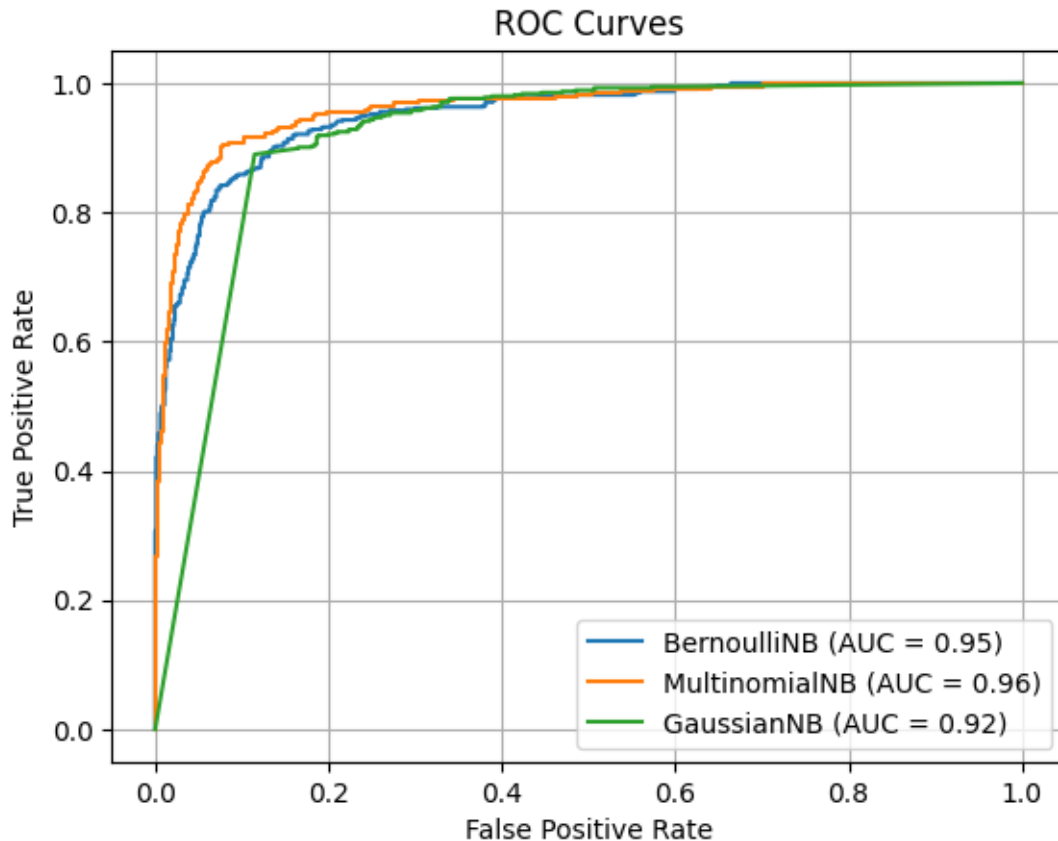
=== GaussianNB ===

Confusion Matrix:

```
[[616 221]
 [ 28 516]]
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.96	0.74	0.83	837
1.0	0.70	0.95	0.81	544
accuracy			0.82	1381
macro avg	0.83	0.84	0.82	1381
weighted avg	0.86	0.82	0.82	1381



=== Comparison Table ===

	Model	Accuracy	Precision	Recall	F1 Score	AUC
0	BernoulliNB	0.889935	0.884314	0.829044	0.855787	0.950023
1	MultinomialNB	0.895004	0.942350	0.781250	0.854271	0.960696
2	GaussianNB	0.819696	0.700136	0.948529	0.805621	0.915675

```
[20]: cv = KFold(n_splits=5, shuffle=True, random_state=42)

# Initialize Multinomial Naive Bayes
mnb = MultinomialNB()

# Perform cross-validation (accuracy as scoring)
scores = cross_val_score(mnb, X_scaled, y, cv=cv, scoring='accuracy')

# Print accuracy for each fold
print(" Multinomial Naive Bayes - 5-Fold Cross-Validation Results:")
for i, score in enumerate(scores, 1):
    print(f"Fold {i} Accuracy: {score:.4f}")
```

```
# Print average and standard deviation
print(f"\nMean Accuracy      : {scores.mean():.4f}")
print(f"Standard Deviation   : {scores.std():.4f}")
```

Multinomial Naive Bayes - 5-Fold Cross-Validation Results:

Fold 1 Accuracy:	0.8719
Fold 2 Accuracy:	0.8935
Fold 3 Accuracy:	0.8891
Fold 4 Accuracy:	0.8913
Fold 5 Accuracy:	0.8859

Mean Accuracy	: 0.8863
Standard Deviation	: 0.0077

5 KNN Code:

knn

August 5, 2025

0.0.1 Importing Required Libraries

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, confusion_matrix, \
    accuracy_score
from sklearn.impute import SimpleImputer
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import matthews_corrcoef
```

0.0.2 Loading dataset

```
[3]: df = pd.read_csv('spambase.csv')
```

0.0.3 Basic info

```
[4]: print("Dataset Info:\n", df.info())
print("\nFirst 5 rows:\n", df.head())
print("\nMissing Values:\n", df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4601 entries, 0 to 4600
```

```
Data columns (total 58 columns):
```

#	Column	Non-Null Count	Dtype
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5	word_freq_over	4601 non-null	float64

6	word_freq_remove	4601 non-null	float64
7	word_freq_internet	4601 non-null	float64
8	word_freq_order	4601 non-null	float64
9	word_freq_mail	4601 non-null	float64
10	word_freq_receive	4601 non-null	float64
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19	word_freq_credit	4601 non-null	float64
20	word_freq_your	4601 non-null	float64
21	word_freq_font	4601 non-null	float64
22	word_freq_000	4601 non-null	float64
23	word_freq_money	4601 non-null	float64
24	word_freq_hp	4601 non-null	float64
25	word_freq_hpl	4601 non-null	float64
26	word_freq_george	4601 non-null	float64
27	word_freq_650	4601 non-null	float64
28	word_freq_lab	4601 non-null	float64
29	word_freq_labs	4601 non-null	float64
30	word_freq_telnet	4601 non-null	float64
31	word_freq_857	4601 non-null	float64
32	word_freq_data	4601 non-null	float64
33	word_freq_415	4601 non-null	float64
34	word_freq_85	4601 non-null	float64
35	word_freq_technology	4601 non-null	float64
36	word_freq_1999	4601 non-null	float64
37	word_freq_parts	4601 non-null	float64
38	word_freq_pm	4601 non-null	float64
39	word_freq_direct	4601 non-null	float64
40	word_freq_cs	4601 non-null	float64
41	word_freq_meeting	4601 non-null	float64
42	word_freq_original	4601 non-null	float64
43	word_freq_project	4601 non-null	float64
44	word_freq_re	4601 non-null	float64
45	word_freq_edu	4601 non-null	float64
46	word_freq_table	4601 non-null	float64
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50	char_freq_%5B	4601 non-null	float64
51	char_freq_%21	4601 non-null	float64
52	char_freq_%24	4601 non-null	float64
53	char_freq_%23	4601 non-null	float64

```

54 capital_run_length_average 4601 non-null float64
55 capital_run_length_longest 4601 non-null int64
56 capital_run_length_total 4601 non-null int64
57 class 4601 non-null int64

```

dtypes: float64(55), int64(3)

memory usage: 2.0 MB

Dataset Info:

None

First 5 rows:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	\
0	0.00	0.64	0.64	0.0	
1	0.21	0.28	0.50	0.0	
2	0.06	0.00	0.71	0.0	
3	0.00	0.00	0.00	0.0	
4	0.00	0.00	0.00	0.0	

	word_freq_our	word_freq_over	word_freq_remove	word_freq_internet	\
0	0.32	0.00	0.00	0.00	
1	0.14	0.28	0.21	0.07	
2	1.23	0.19	0.19	0.12	
3	0.63	0.00	0.31	0.63	
4	0.63	0.00	0.31	0.63	

	word_freq_order	word_freq_mail	...	char_freq_%3B	char_freq_%28	\
0	0.00	0.00	...	0.00	0.000	
1	0.00	0.94	...	0.00	0.132	
2	0.64	0.25	...	0.01	0.143	
3	0.31	0.63	...	0.00	0.137	
4	0.31	0.63	...	0.00	0.135	

	char_freq_%5B	char_freq_%21	char_freq_%24	char_freq_%23	\
0	0.0	0.778	0.000	0.000	
1	0.0	0.372	0.180	0.048	
2	0.0	0.276	0.184	0.010	
3	0.0	0.137	0.000	0.000	
4	0.0	0.135	0.000	0.000	

	capital_run_length_average	capital_run_length_longest	\
0	3.756	61	
1	5.114	101	
2	9.821	485	
3	3.537	40	
4	3.537	40	

	capital_run_length_total	class
0	278	1
1	1028	1

2	2259	1
3	191	1
4	191	1

[5 rows x 58 columns]

Missing Values:

word_freq_make	0
word_freq_address	0
word_freq_all	0
word_freq_3d	0
word_freq_our	0
word_freq_over	0
word_freq_remove	0
word_freq_internet	0
word_freq_order	0
word_freq_mail	0
word_freq_receive	0
word_freq_will	0
word_freq_people	0
word_freq_report	0
word_freq_addresses	0
word_freq_free	0
word_freq_business	0
word_freq_email	0
word_freq_you	0
word_freq_credit	0
word_freq_your	0
word_freq_font	0
word_freq_000	0
word_freq_money	0
word_freq_hp	0
word_freq_hpl	0
word_freq_george	0
word_freq_650	0
word_freq_lab	0
word_freq_labs	0
word_freq_telnet	0
word_freq_857	0
word_freq_data	0
word_freq_415	0
word_freq_85	0
word_freq_technology	0
word_freq_1999	0
word_freq_parts	0
word_freq_pm	0
word_freq_direct	0
word_freq_cs	0

```

word_freq_meeting      0
word_freq_original     0
word_freq_project      0
word_freq_re           0
word_freq_edu          0
word_freq_table        0
word_freq_conference   0
char_freq_%3B          0
char_freq_%28          0
char_freq_%5B          0
char_freq_%21          0
char_freq_%24          0
char_freq_%23          0
capital_run_length_average 0
capital_run_length_longest 0
capital_run_length_total 0
class                  0
dtype: int64

```

0.0.4 Handling missing values

```

[5]: imputer = SimpleImputer(strategy='mean')
     df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)

```

0.0.5 Splitting of feature and target

```

[6]: X = df_imputed.drop('class', axis=1)
     y = df_imputed['class']

```

0.0.6 Checking Distribution

```

[7]: X = df.drop('class', axis=1)  # Assuming 'class' is the target

# Choose a few features to visualize
sample_features = X.columns[:5]  # First 5 features

# Plot histograms with Gaussian curve overlay
plt.figure(figsize=(15, 10))
for i, feature in enumerate(sample_features):
    plt.subplot(3, 2, i + 1)
    data = X[feature]

    # Plot histogram
    count, bins, ignored = plt.hist(data, bins=30, density=True, alpha=0.6,
    ↪ color='skyblue', edgecolor='black')

    # Plot normal distribution curve

```

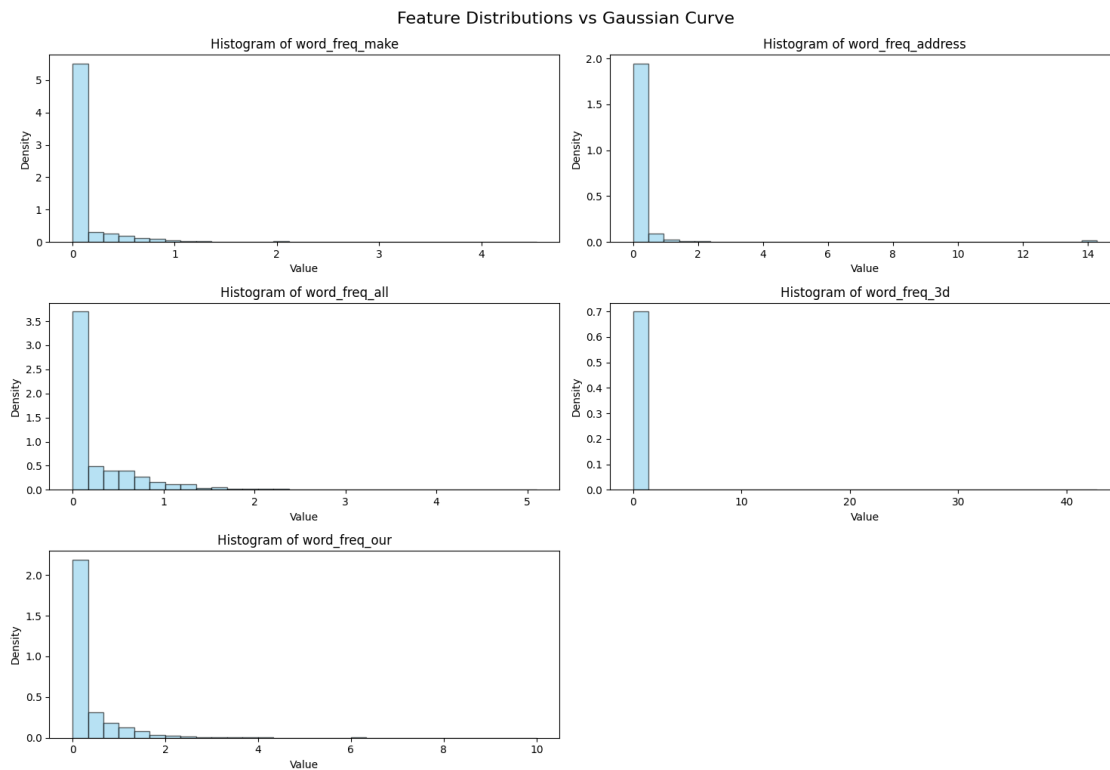
```

'''mu, std = data.mean(), data.std()
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = 1 / (std * np.sqrt(2 * np.pi)) * np.exp(-(x - mu)**2 / (2 * std**2))
plt.plot(x, p, 'r', linewidth=2)'''

plt.title(f'Histogram of {feature}')
plt.xlabel('Value')
plt.ylabel('Density')

plt.tight_layout()
plt.suptitle('Feature Distributions vs Gaussian Curve', fontsize=16, y=1.02)
plt.show()

```



0.0.7 Applying min max scaling

```

[8]: scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

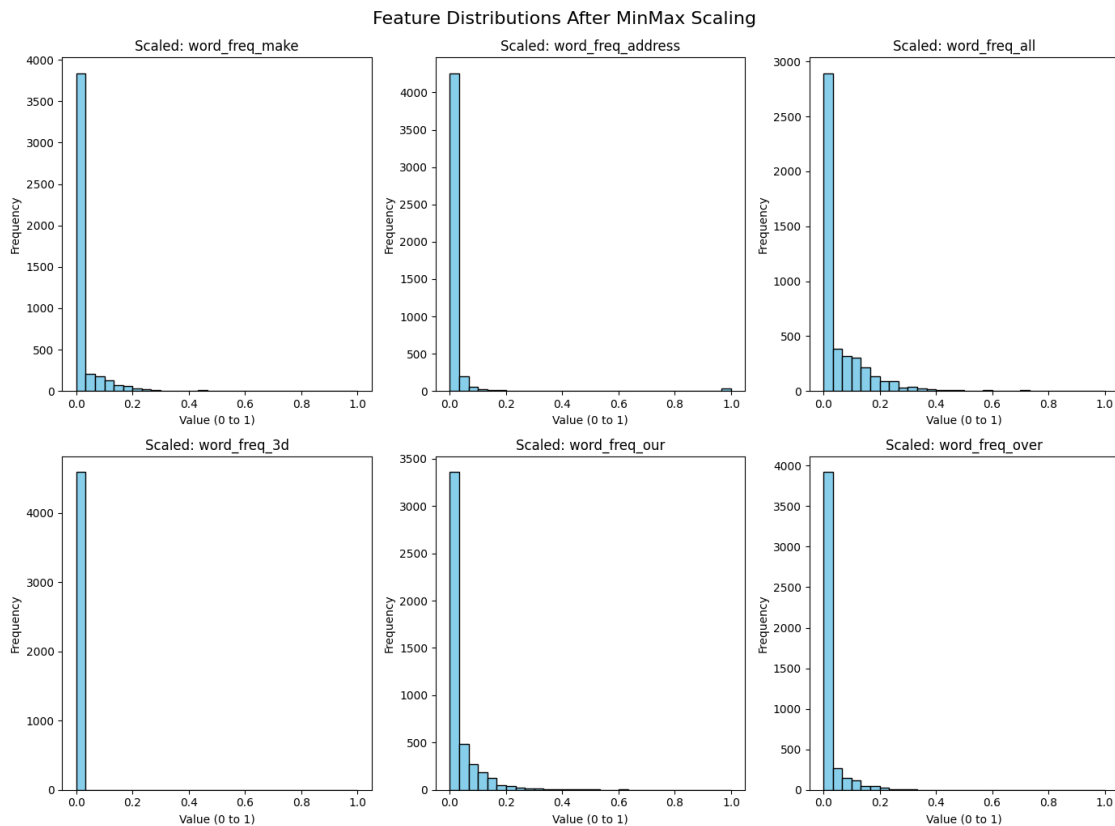
```

##Plots

0.0.8 Histogram

```
[9]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))
for i in range(6): # first 6 features as an example
    plt.subplot(2, 3, i + 1)
    plt.hist(X_scaled[:, i], bins=30, color='skyblue', edgecolor='black')
    plt.title(f'Scaled: {X.columns[i]}')
    plt.xlabel('Value (0 to 1)')
    plt.ylabel('Frequency')
plt.tight_layout()
plt.suptitle('Feature Distributions After MinMax Scaling', fontsize=16, y=1.02)
plt.show()
```

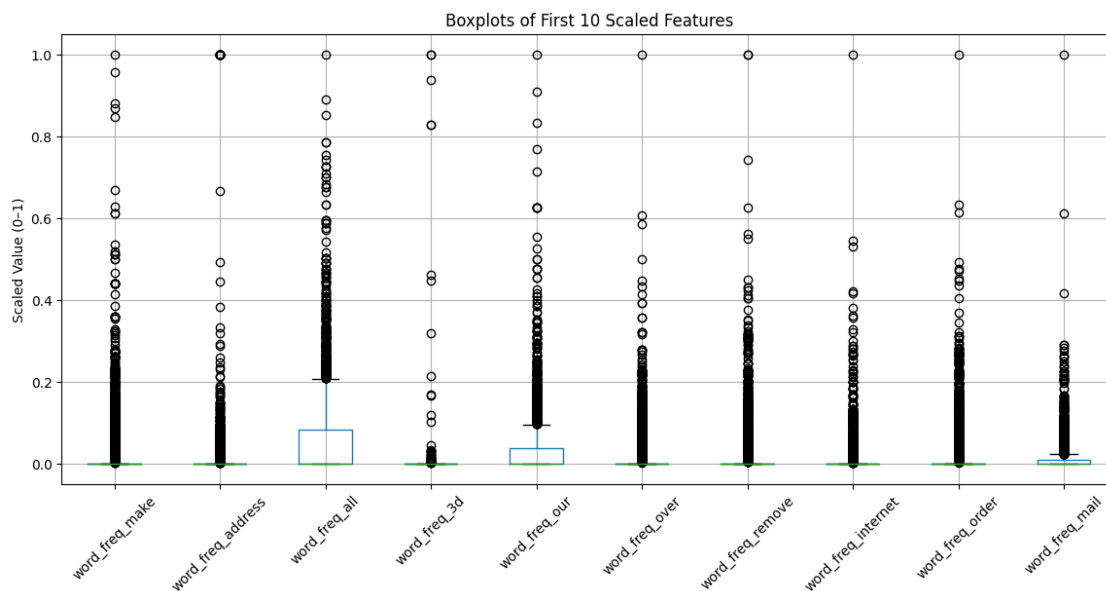


0.0.9 Boxplot

```
[10]: import pandas as pd

# Convert scaled array back to DataFrame for easier plotting
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)
```

```
plt.figure(figsize=(14, 6))
X_scaled_df.iloc[:, :10].boxplot(rot=45)
plt.title('Boxplots of First 10 Scaled Features')
plt.ylabel('Scaled Value (0-1)')
plt.grid(True)
plt.show()
```



0.0.10 Correlation HeatMap

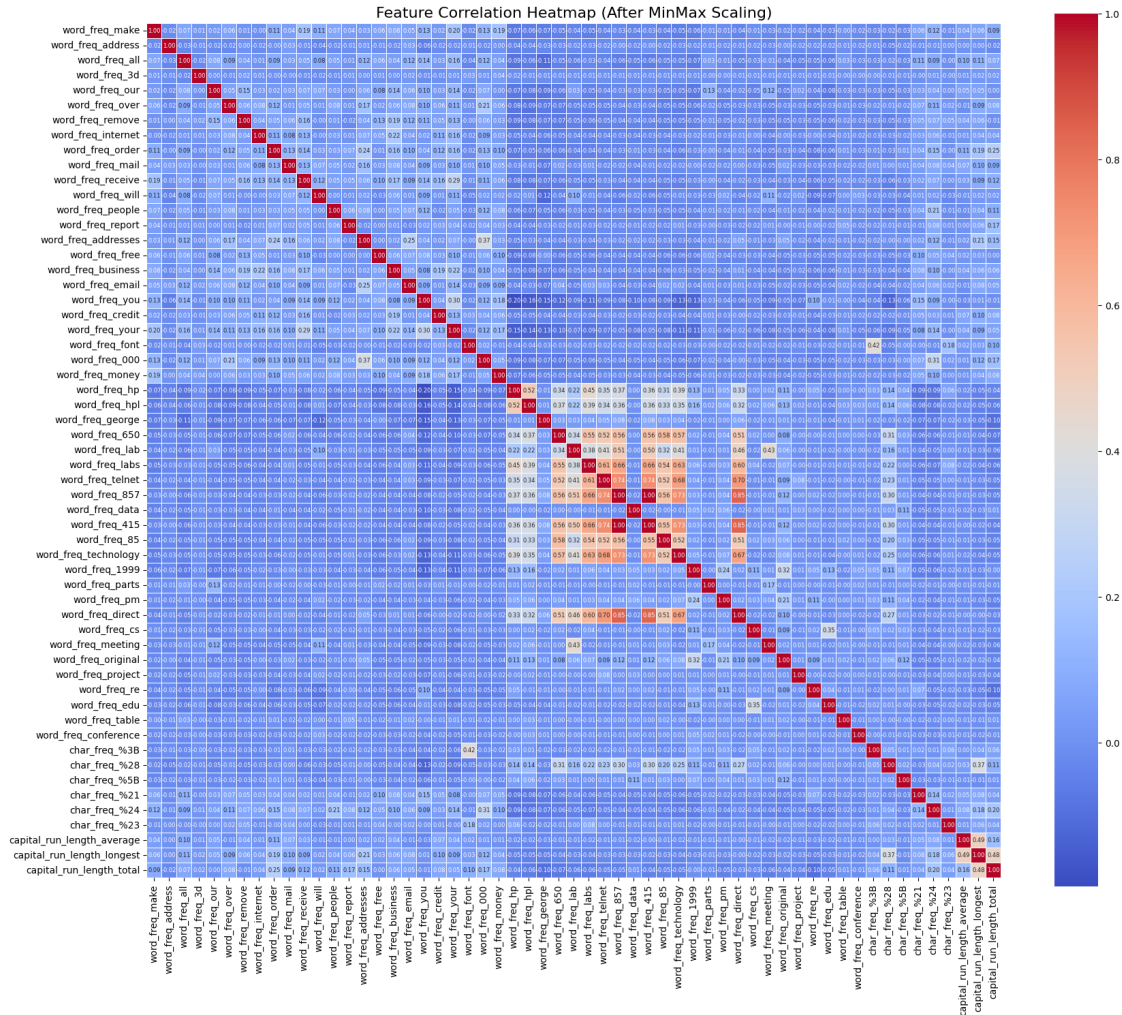
```
[11]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Assuming X_scaled_df is your scaled DataFrame
corr_matrix = X_scaled_df.corr()

plt.figure(figsize=(18, 15)) # Bigger figure
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
    square=True,
    annot=True,                # Show values inside squares
    fmt=".2f",                 # Format to 2 decimal places
    linewidths=0.5,           # Thin grid lines
    annot_kws={"size": 6}     # Smaller font size
)
```



```
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.title("Feature Correlation Heatmap (After MinMax Scaling)", fontsize=16)
plt.tight_layout()
plt.show()
```



0.0.11 Choosing best k value using k-fold cross validation

```
[12]: # Step 1: 5-Fold Stratified CV Setup
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Step 2: Try different k values
k_range = range(1, 21)
mean accuracies = []
```

```

print("Cross-Validation Accuracies for Different k:")
print("-" * 45)

for k in k_range:
    model = KNeighborsClassifier(n_neighbors=k)
    scores = cross_val_score(model, X_scaled, y, cv=cv, scoring='accuracy')
    mean_accuracies.append(scores.mean())

    print(f"k = {k:2d} -> Fold Accuracies: {np.round(scores, 4)} -> Mean_
↳Accuracy: {scores.mean():.4f}")

# Step 3: Plot Mean Accuracy vs. k
plt.figure(figsize=(10, 5))
plt.plot(k_range, mean_accuracies, marker='o', linestyle='--', color='blue')
plt.title('Mean Cross-Validation Accuracy vs. k')
plt.xlabel('Number of Neighbors (k)')
plt.ylabel('Mean Accuracy')
plt.xticks(k_range)
plt.grid(True)
plt.show()

# Step 4: Best k
best_k = k_range[np.argmax(mean_accuracies)]
print(f"\n Best k based on mean CV accuracy: {best_k}")

```

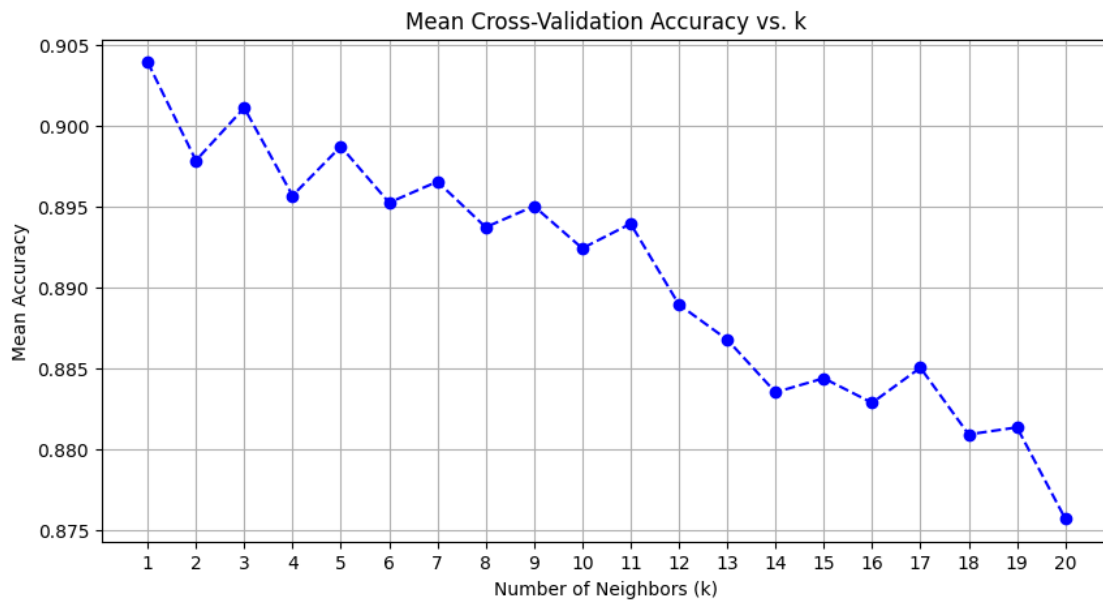
Cross-Validation Accuracies for Different k:

```

-----
k = 1 -> Fold Accuracies: [0.9034 0.9076 0.9152 0.9    0.8935] -> Mean
Accuracy: 0.9039
k = 2 -> Fold Accuracies: [0.8914 0.8924 0.9076 0.8989 0.8989] -> Mean
Accuracy: 0.8978
k = 3 -> Fold Accuracies: [0.8947 0.8989 0.9087 0.9087 0.8946] -> Mean
Accuracy: 0.9011
k = 4 -> Fold Accuracies: [0.8947 0.8902 0.9022 0.9043 0.887 ] -> Mean
Accuracy: 0.8957
k = 5 -> Fold Accuracies: [0.8969 0.8946 0.9011 0.9054 0.8957] -> Mean
Accuracy: 0.8987
k = 6 -> Fold Accuracies: [0.9001 0.8902 0.8967 0.9011 0.888 ] -> Mean
Accuracy: 0.8952
k = 7 -> Fold Accuracies: [0.899  0.8978 0.8989 0.8946 0.8924] -> Mean
Accuracy: 0.8965
k = 8 -> Fold Accuracies: [0.8914 0.8891 0.9033 0.8946 0.8902] -> Mean
Accuracy: 0.8937
k = 9 -> Fold Accuracies: [0.8936 0.8913 0.9065 0.8978 0.8859] -> Mean
Accuracy: 0.8950
k = 10 -> Fold Accuracies: [0.8947 0.888  0.9033 0.888  0.888 ] -> Mean
Accuracy: 0.8924

```

k = 11 -> Fold Accuracies: [0.8958 0.8924 0.9054 0.8957 0.8804] -> Mean Accuracy: 0.8939
 k = 12 -> Fold Accuracies: [0.8925 0.8826 0.8967 0.8924 0.8804] -> Mean Accuracy: 0.8889
 k = 13 -> Fold Accuracies: [0.8893 0.8826 0.8957 0.8935 0.8728] -> Mean Accuracy: 0.8868
 k = 14 -> Fold Accuracies: [0.8914 0.8783 0.8859 0.8891 0.8728] -> Mean Accuracy: 0.8835
 k = 15 -> Fold Accuracies: [0.8882 0.8826 0.8946 0.8913 0.8652] -> Mean Accuracy: 0.8844
 k = 16 -> Fold Accuracies: [0.8849 0.8793 0.8891 0.887 0.8739] -> Mean Accuracy: 0.8829
 k = 17 -> Fold Accuracies: [0.8893 0.887 0.8902 0.8946 0.8641] -> Mean Accuracy: 0.8850
 k = 18 -> Fold Accuracies: [0.886 0.8826 0.8902 0.8826 0.863] -> Mean Accuracy: 0.8809
 k = 19 -> Fold Accuracies: [0.8795 0.8837 0.8935 0.8902 0.8598] -> Mean Accuracy: 0.8813
 k = 20 -> Fold Accuracies: [0.8773 0.8761 0.8902 0.8837 0.8511] -> Mean Accuracy: 0.8757



Best k based on mean CV accuracy: 1

0.0.12 Model Training and Evaluation

```
[21]: from sklearn.metrics import (
        accuracy_score, precision_score, recall_score,
        f1_score, confusion_matrix, roc_curve, auc,
        classification_report
    )

    # Split the data again (or reuse your earlier split)
    from sklearn.model_selection import train_test_split
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled, y, test_size=0.3, stratify=y, random_state=42
    )

    # Store results
    knn_results = []

    # Plot setup for ROC
    plt.figure(figsize=(8, 6))

    # Loop through different k values
    for k in [1, 3, 5, 7]:
        model = KNeighborsClassifier(n_neighbors=k)
        model.fit(X_train, y_train)

        y_pred = model.predict(X_test)
        y_prob = model.predict_proba(X_test)[:, 1]

        acc = accuracy_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred, zero_division=0)
        rec = recall_score(y_test, y_pred, zero_division=0)
        f1 = f1_score(y_test, y_pred, zero_division=0)
        mcc = matthews_corrcoef(y_test, y_pred)

        fpr, tpr, _ = roc_curve(y_test, y_prob)
        roc_auc = auc(fpr, tpr)

        # Plot ROC
        plt.plot(fpr, tpr, label=f'k={k} (AUC={roc_auc:.2f})')

    print(f"\n KNN Evaluation for k = {k}")
    print(f"Accuracy : {acc:.4f}")
    print(f"Precision: {prec:.4f}")
    print(f"Recall    : {rec:.4f}")
    print(f"F1 Score  : {f1:.4f}")
    print(f"MCC      : {mcc:.4f}")
    print("\nClassification Report:")
```

```

print(classification_report(y_test, y_pred, zero_division=0))

knn_results.append({
    'k': k,
    'Accuracy': acc,
    'Precision': prec,
    'Recall': rec,
    'F1 Score': f1,
    'MCC': mcc,
    'AUC': roc_auc
})

# Final ROC Curve Plot
plt.title("ROC Curves for Different K Values")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.grid(True)
plt.legend()
plt.show()

# Comparison Table
knn_df = pd.DataFrame(knn_results)
print("\n Comparison Table:")
print(knn_df)

```

KNN Evaluation for k = 1

Accuracy : 0.8899
Precision: 0.8551
Recall : 0.8676
F1 Score : 0.8613
MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381

KNN Evaluation for k = 3

Accuracy : 0.8892
Precision: 0.8710
Recall : 0.8438

F1 Score : 0.8571
MCC : 0.7670

Classification Report:

	precision	recall	f1-score	support
0.0	0.90	0.92	0.91	837
1.0	0.87	0.84	0.86	544
accuracy			0.89	1381
macro avg	0.89	0.88	0.88	1381
weighted avg	0.89	0.89	0.89	1381

KNN Evaluation for k = 5

Accuracy : 0.8993
Precision: 0.8828
Recall : 0.8585
F1 Score : 0.8705
MCC : 0.7884

Classification Report:

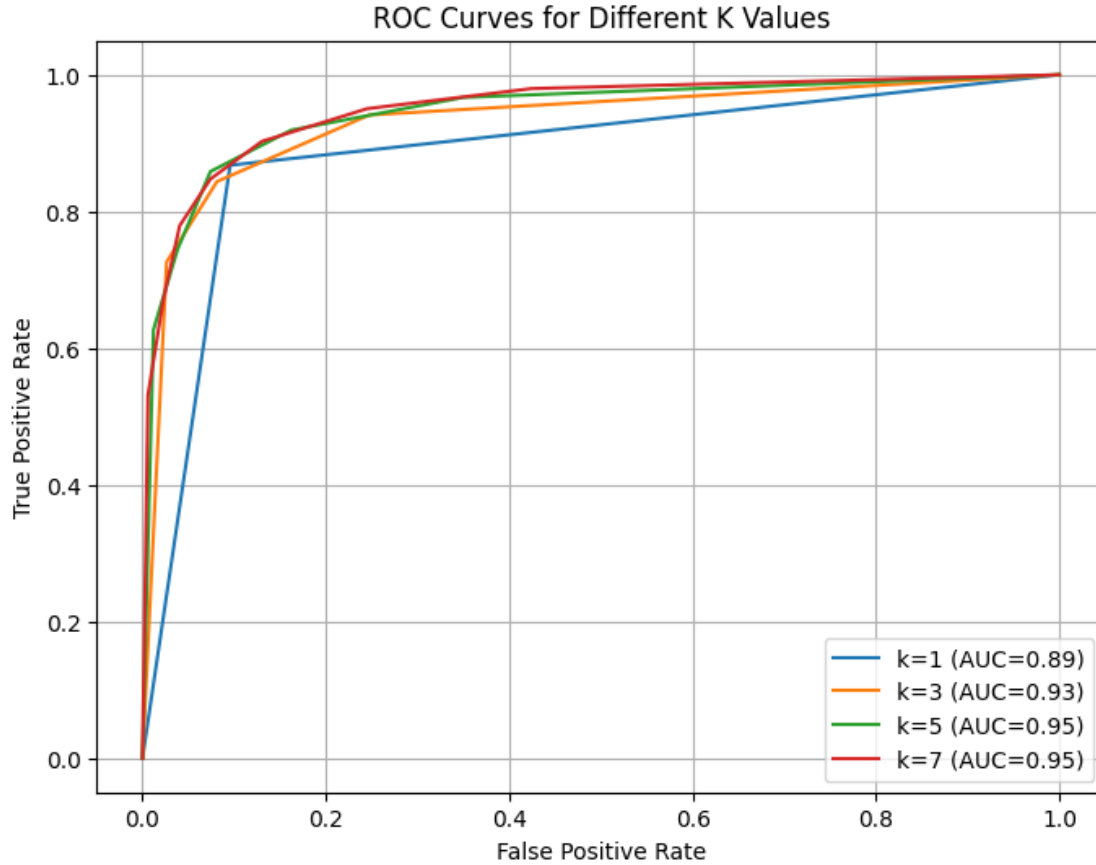
	precision	recall	f1-score	support
0.0	0.91	0.93	0.92	837
1.0	0.88	0.86	0.87	544
accuracy			0.90	1381
macro avg	0.90	0.89	0.89	1381
weighted avg	0.90	0.90	0.90	1381

KNN Evaluation for k = 7

Accuracy : 0.8950
Precision: 0.8815
Recall : 0.8474
F1 Score : 0.8641
MCC : 0.7790

Classification Report:

	precision	recall	f1-score	support
0.0	0.90	0.93	0.91	837
1.0	0.88	0.85	0.86	544
accuracy			0.90	1381
macro avg	0.89	0.89	0.89	1381
weighted avg	0.89	0.90	0.89	1381



Comparison Table:

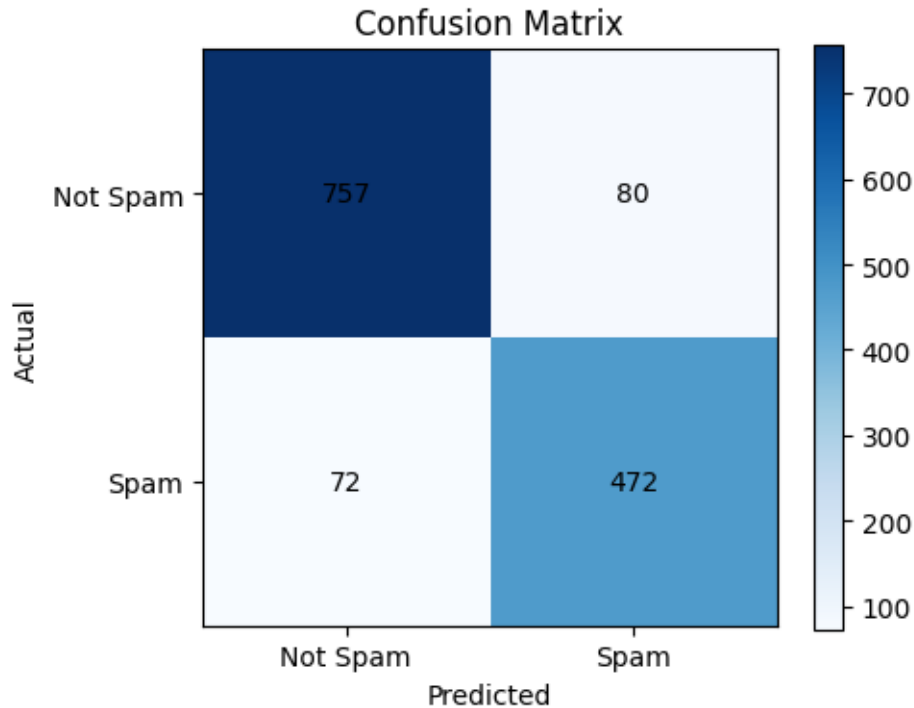
	k	Accuracy	Precision	Recall	F1 Score	MCC	AUC
0	1	0.889935	0.855072	0.867647	0.861314	0.770142	0.886034
1	3	0.889211	0.870968	0.843750	0.857143	0.766959	0.931442
2	5	0.899348	0.882798	0.858456	0.870457	0.788392	0.945145
3	7	0.895004	0.881453	0.847426	0.864105	0.779014	0.950240

```
[15]: conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(5, 4))
plt.imshow(conf_matrix, cmap='Blues')
plt.title('Confusion Matrix')
plt.colorbar()
plt.xticks([0, 1], ['Not Spam', 'Spam'])
plt.yticks([0, 1], ['Not Spam', 'Spam'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
for i in range(2):
```

```

    for j in range(2):
        plt.text(j, i, conf_matrix[i, j], ha='center', va='center',
        ↪color='black')
plt.tight_layout()
plt.show()

```



0.0.13 KD Tree and Ball Tree

```

[17]: # Use KDTree
model_kd = KNeighborsClassifier(n_neighbors=best_k, algorithm='kd_tree')
model_kd.fit(X_train, y_train)

# Use BallTree
model_ball = KNeighborsClassifier(n_neighbors=best_k, algorithm='ball_tree')
model_ball.fit(X_train, y_train)

```

```

[17]: KNeighborsClassifier(algorithm='ball_tree', n_neighbors=1)

```

```

[18]: from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, roc_curve, auc,
    classification_report, matthews_corrcoef
)

```



```

)
import matplotlib.pyplot as plt

# Helper function to evaluate a model
def evaluate_knn_model(name, model, X_train, X_test, y_train, y_test):
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    mcc = matthews_corrcoef(y_test, y_pred)
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    roc_auc = auc(fpr, tpr)

    print(f"\n {name} Performance Metrics:")
    print(f"Accuracy : {acc:.4f}")
    print(f"Precision: {prec:.4f}")
    print(f"Recall    : {rec:.4f}")
    print(f"F1 Score  : {f1:.4f}")
    print(f"MCC      : {mcc:.4f}")
    print("\nClassification Report:\n", classification_report(y_test, y_pred))

    # Confusion Matrix
    conf = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(5, 4))
    plt.imshow(conf, cmap='Blues')
    plt.title(f'{name} Confusion Matrix')
    plt.colorbar()
    plt.xticks([0, 1], ['Not Spam', 'Spam'])
    plt.yticks([0, 1], ['Not Spam', 'Spam'])
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    for i in range(2):
        for j in range(2):
            plt.text(j, i, conf[i, j], ha='center', va='center')
    plt.tight_layout()
    plt.show()

    # ROC Curve
    plt.figure(figsize=(6, 5))
    plt.plot(fpr, tpr, label=f'{name} (AUC = {roc_auc:.4f})')
    plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')

```

```

plt.title(f'{name} ROC Curve')
plt.legend()
plt.grid(True)
plt.show()

# Train-test split assumed done:
# X_train, X_test, y_train, y_test already available
# best_k is selected

# Evaluate KDTree
knn_kd = KNeighborsClassifier(n_neighbors=best_k, algorithm='kd_tree')
evaluate_knn_model("KDTree", knn_kd, X_train, X_test, y_train, y_test)

# Evaluate BallTree
knn_ball = KNeighborsClassifier(n_neighbors=best_k, algorithm='ball_tree')
evaluate_knn_model("BallTree", knn_ball, X_train, X_test, y_train, y_test)

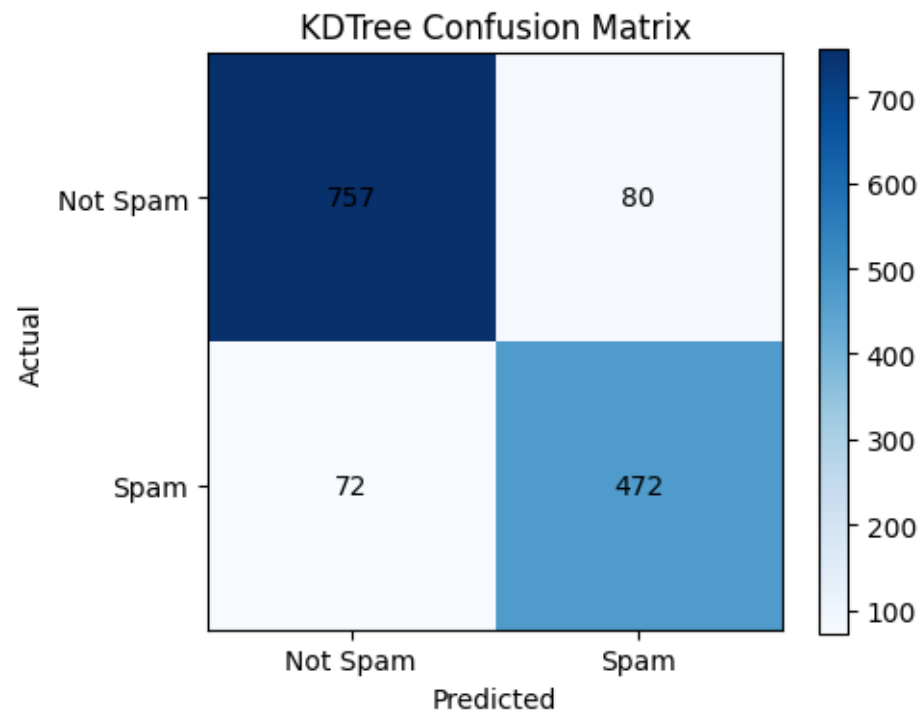
```

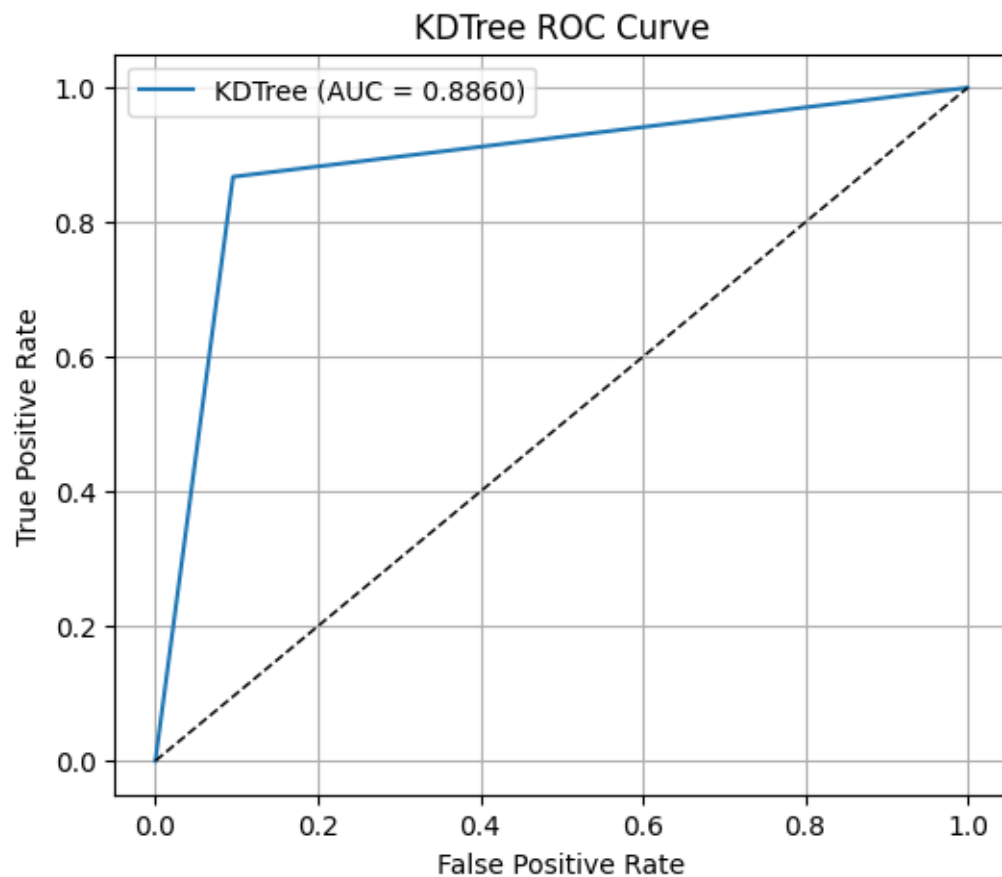
KDTree Performance Metrics:

Accuracy : 0.8899
 Precision: 0.8551
 Recall : 0.8676
 F1 Score : 0.8613
 MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381



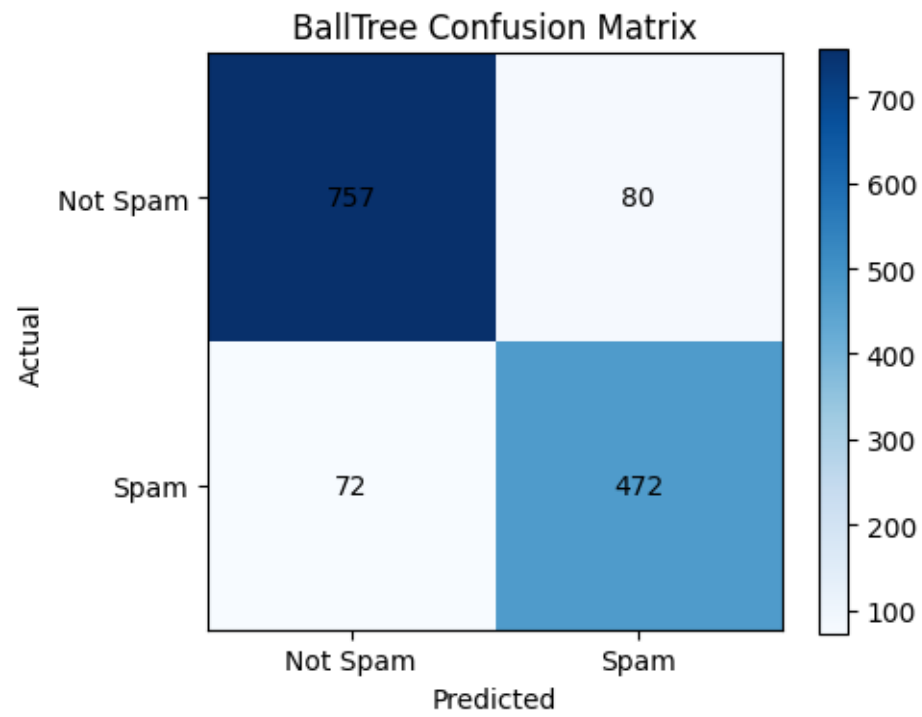


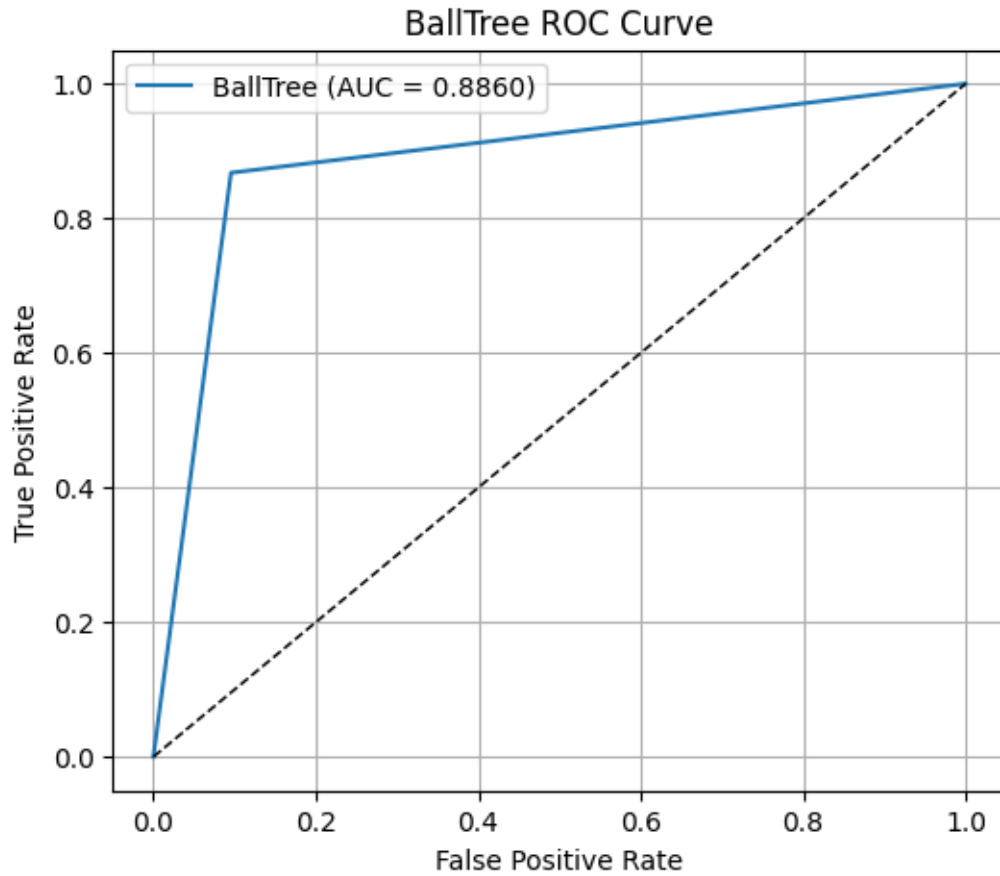
BallTree Performance Metrics:

Accuracy : 0.8899
Precision: 0.8551
Recall : 0.8676
F1 Score : 0.8613
MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381





```
[26]: import time

for algo in ['kd_tree', 'ball_tree', 'brute']:
    model = KNeighborsClassifier(n_neighbors=best_k, algorithm=algo)
    start = time.time()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    duration = time.time() - start
    acc = accuracy_score(y_test, y_pred)
    print(f"{algo:10} → Accuracy: {acc:.4f}, Time: {duration:.4f} sec")
```

```
kd_tree    → Accuracy: 0.8899, Time: 0.4470 sec
ball_tree  → Accuracy: 0.8899, Time: 0.4058 sec
brute      → Accuracy: 0.8899, Time: 0.0192 sec
```

6 SVM Code:

SVM

September 2, 2025

0.0.1 Importing Required Libraries

```
[ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler # or MinMaxScaler, if used
from sklearn.preprocessing import LabelEncoder # if categorical labels
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, \
    classification_report
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.model_selection import cross_val_score, KFold, GridSearchCV
from sklearn.model_selection import cross_val_score, StratifiedKFold
```

0.0.2 Loading dataset

```
[ ]: df = pd.read_csv('spambase.csv')
```

0.0.3 Basic info

```
[ ]: print("Dataset Info:\n", df.info())
print("\nFirst 5 rows:\n", df.head())
print("\nMissing Values:\n", df.isnull().sum())
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 4601 entries, 0 to 4600

Data columns (total 58 columns):

#	Column	Non-Null Count	Dtype
0	word_freq_make	4601 non-null	float64
1	word_freq_address	4601 non-null	float64
2	word_freq_all	4601 non-null	float64
3	word_freq_3d	4601 non-null	float64
4	word_freq_our	4601 non-null	float64
5	word_freq_over	4601 non-null	float64

6	word_freq_remove	4601 non-null	float64
7	word_freq_internet	4601 non-null	float64
8	word_freq_order	4601 non-null	float64
9	word_freq_mail	4601 non-null	float64
10	word_freq_receive	4601 non-null	float64
11	word_freq_will	4601 non-null	float64
12	word_freq_people	4601 non-null	float64
13	word_freq_report	4601 non-null	float64
14	word_freq_addresses	4601 non-null	float64
15	word_freq_free	4601 non-null	float64
16	word_freq_business	4601 non-null	float64
17	word_freq_email	4601 non-null	float64
18	word_freq_you	4601 non-null	float64
19	word_freq_credit	4601 non-null	float64
20	word_freq_your	4601 non-null	float64
21	word_freq_font	4601 non-null	float64
22	word_freq_000	4601 non-null	float64
23	word_freq_money	4601 non-null	float64
24	word_freq_hp	4601 non-null	float64
25	word_freq_hpl	4601 non-null	float64
26	word_freq_george	4601 non-null	float64
27	word_freq_650	4601 non-null	float64
28	word_freq_lab	4601 non-null	float64
29	word_freq_labs	4601 non-null	float64
30	word_freq_telnet	4601 non-null	float64
31	word_freq_857	4601 non-null	float64
32	word_freq_data	4601 non-null	float64
33	word_freq_415	4601 non-null	float64
34	word_freq_85	4601 non-null	float64
35	word_freq_technology	4601 non-null	float64
36	word_freq_1999	4601 non-null	float64
37	word_freq_parts	4601 non-null	float64
38	word_freq_pm	4601 non-null	float64
39	word_freq_direct	4601 non-null	float64
40	word_freq_cs	4601 non-null	float64
41	word_freq_meeting	4601 non-null	float64
42	word_freq_original	4601 non-null	float64
43	word_freq_project	4601 non-null	float64
44	word_freq_re	4601 non-null	float64
45	word_freq_edu	4601 non-null	float64
46	word_freq_table	4601 non-null	float64
47	word_freq_conference	4601 non-null	float64
48	char_freq_%3B	4601 non-null	float64
49	char_freq_%28	4601 non-null	float64
50	char_freq_%5B	4601 non-null	float64
51	char_freq_%21	4601 non-null	float64
52	char_freq_%24	4601 non-null	float64
53	char_freq_%23	4601 non-null	float64

```

54 capital_run_length_average 4601 non-null float64
55 capital_run_length_longest 4601 non-null int64
56 capital_run_length_total 4601 non-null int64
57 class 4601 non-null int64

```

dtypes: float64(55), int64(3)

memory usage: 2.0 MB

Dataset Info:

None

First 5 rows:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	\
0	0.00	0.64	0.64	0.0	
1	0.21	0.28	0.50	0.0	
2	0.06	0.00	0.71	0.0	
3	0.00	0.00	0.00	0.0	
4	0.00	0.00	0.00	0.0	

	word_freq_our	word_freq_over	word_freq_remove	word_freq_internet	\
0	0.32	0.00	0.00	0.00	
1	0.14	0.28	0.21	0.07	
2	1.23	0.19	0.19	0.12	
3	0.63	0.00	0.31	0.63	
4	0.63	0.00	0.31	0.63	

	word_freq_order	word_freq_mail	...	char_freq_%3B	char_freq_%28	\
0	0.00	0.00	...	0.00	0.000	
1	0.00	0.94	...	0.00	0.132	
2	0.64	0.25	...	0.01	0.143	
3	0.31	0.63	...	0.00	0.137	
4	0.31	0.63	...	0.00	0.135	

	char_freq_%5B	char_freq_%21	char_freq_%24	char_freq_%23	\
0	0.0	0.778	0.000	0.000	
1	0.0	0.372	0.180	0.048	
2	0.0	0.276	0.184	0.010	
3	0.0	0.137	0.000	0.000	
4	0.0	0.135	0.000	0.000	

	capital_run_length_average	capital_run_length_longest	\
0	3.756	61	
1	5.114	101	
2	9.821	485	
3	3.537	40	
4	3.537	40	

	capital_run_length_total	class
0	278	1
1	1028	1

2	2259	1
3	191	1
4	191	1

[5 rows x 58 columns]

Missing Values:

word_freq_make	0
word_freq_address	0
word_freq_all	0
word_freq_3d	0
word_freq_our	0
word_freq_over	0
word_freq_remove	0
word_freq_internet	0
word_freq_order	0
word_freq_mail	0
word_freq_receive	0
word_freq_will	0
word_freq_people	0
word_freq_report	0
word_freq_addresses	0
word_freq_free	0
word_freq_business	0
word_freq_email	0
word_freq_you	0
word_freq_credit	0
word_freq_your	0
word_freq_font	0
word_freq_000	0
word_freq_money	0
word_freq_hp	0
word_freq_hpl	0
word_freq_george	0
word_freq_650	0
word_freq_lab	0
word_freq_labs	0
word_freq_telnet	0
word_freq_857	0
word_freq_data	0
word_freq_415	0
word_freq_85	0
word_freq_technology	0
word_freq_1999	0
word_freq_parts	0
word_freq_pm	0
word_freq_direct	0
word_freq_cs	0

```

word_freq_meeting      0
word_freq_original     0
word_freq_project      0
word_freq_re           0
word_freq_edu          0
word_freq_table        0
word_freq_conference   0
char_freq_%3B          0
char_freq_%28          0
char_freq_%5B          0
char_freq_%21          0
char_freq_%24          0
char_freq_%23          0
capital_run_length_average 0
capital_run_length_longest 0
capital_run_length_total 0
class                  0
dtype: int64

```

0.0.4 Handling missing values

```
[ ]: imputer = SimpleImputer(strategy='mean')
df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)
```

0.0.5 Splitting of feature and target

```
[ ]: X = df_imputed.drop('class', axis=1)
y = df_imputed['class']
```

0.0.6 Checking Distribution

```
[ ]: X = df.drop('class', axis=1) # Assuming 'class' is the target

# Choose a few features to visualize
sample_features = X.columns[:5] # First 5 features

# Plot histograms with Gaussian curve overlay
plt.figure(figsize=(15, 10))
for i, feature in enumerate(sample_features):
    plt.subplot(3, 2, i + 1)
    data = X[feature]

    # Plot histogram
    count, bins, ignored = plt.hist(data, bins=30, density=True, alpha=0.6,
    color='skyblue', edgecolor='black')

    # Plot normal distribution curve
```

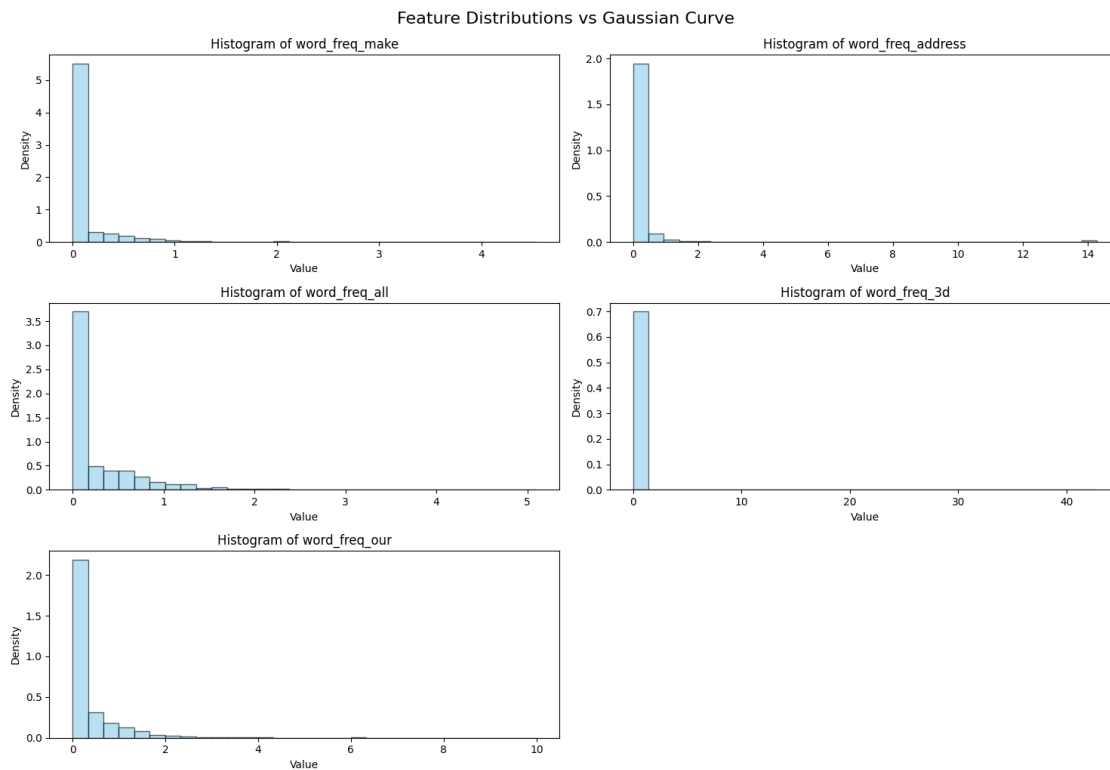
```

'''mu, std = data.mean(), data.std()
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = 1 / (std * np.sqrt(2 * np.pi)) * np.exp(-(x - mu)**2 / (2 * std**2))
plt.plot(x, p, 'r', linewidth=2)'''

plt.title(f'Histogram of {feature}')
plt.xlabel('Value')
plt.ylabel('Density')

plt.tight_layout()
plt.suptitle('Feature Distributions vs Gaussian Curve', fontsize=16, y=1.02)
plt.show()

```



0.0.7 Applying min max scaling

```

[ ]: scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

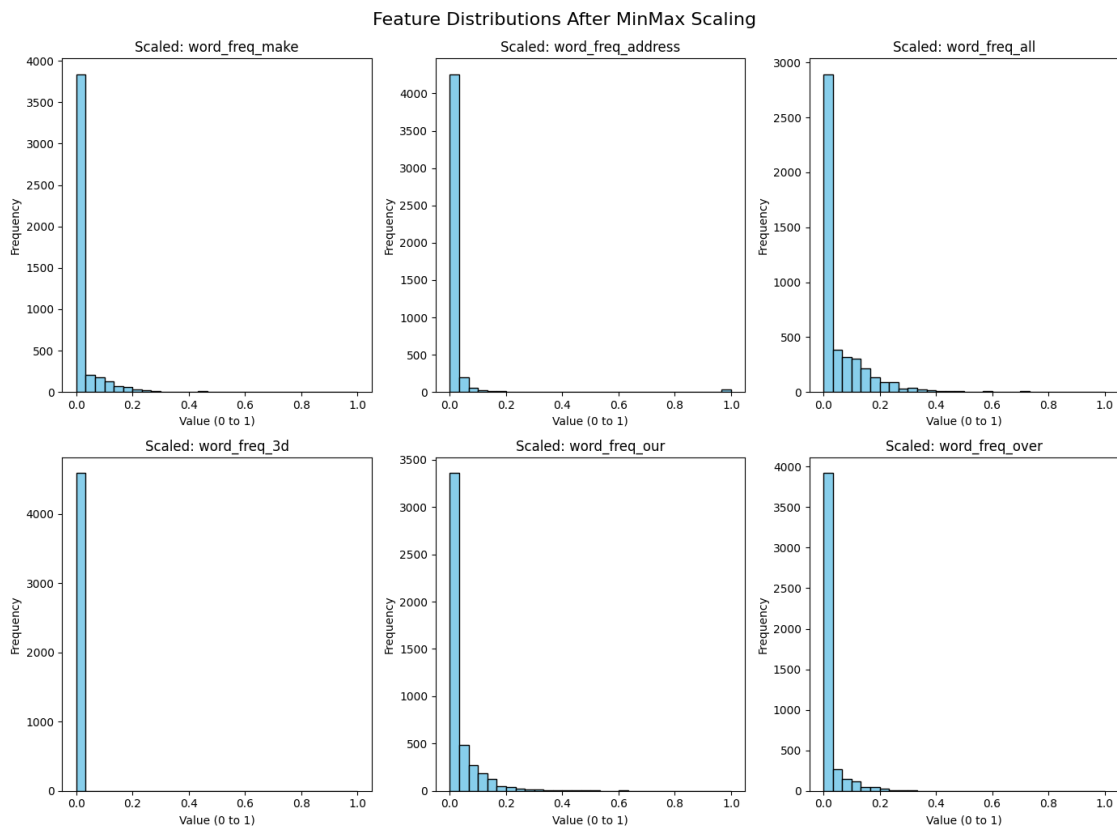
```

0.1 Plots

0.1.1 Histogram

```
[ ]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))
for i in range(6): # first 6 features as an example
    plt.subplot(2, 3, i + 1)
    plt.hist(X_scaled[:, i], bins=30, color='skyblue', edgecolor='black')
    plt.title(f'Scaled: {X.columns[i]}')
    plt.xlabel('Value (0 to 1)')
    plt.ylabel('Frequency')
plt.tight_layout()
plt.suptitle('Feature Distributions After MinMax Scaling', fontsize=16, y=1.02)
plt.show()
```

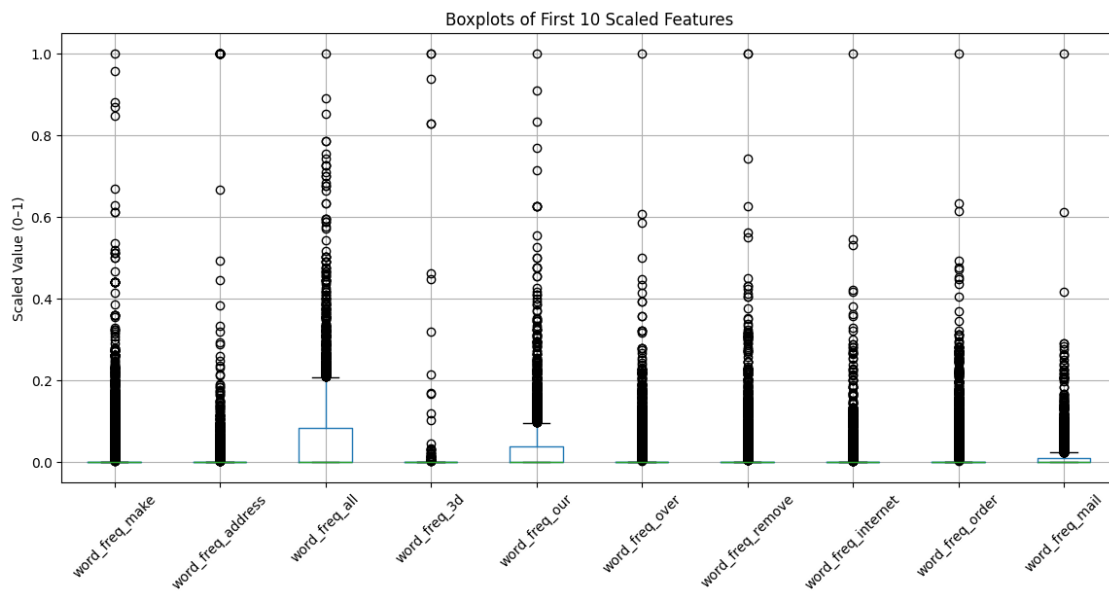


0.1.2 Boxplot

```
[ ]: import pandas as pd

# Convert scaled array back to DataFrame for easier plotting
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)

plt.figure(figsize=(14, 6))
X_scaled_df.iloc[:, :10].boxplot(rot=45)
plt.title('Boxplots of First 10 Scaled Features')
plt.ylabel('Scaled Value (0-1)')
plt.grid(True)
plt.show()
```



0.1.3 Correlation HeatMap

```
[ ]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Assuming X_scaled_df is your scaled DataFrame
corr_matrix = X_scaled_df.corr()

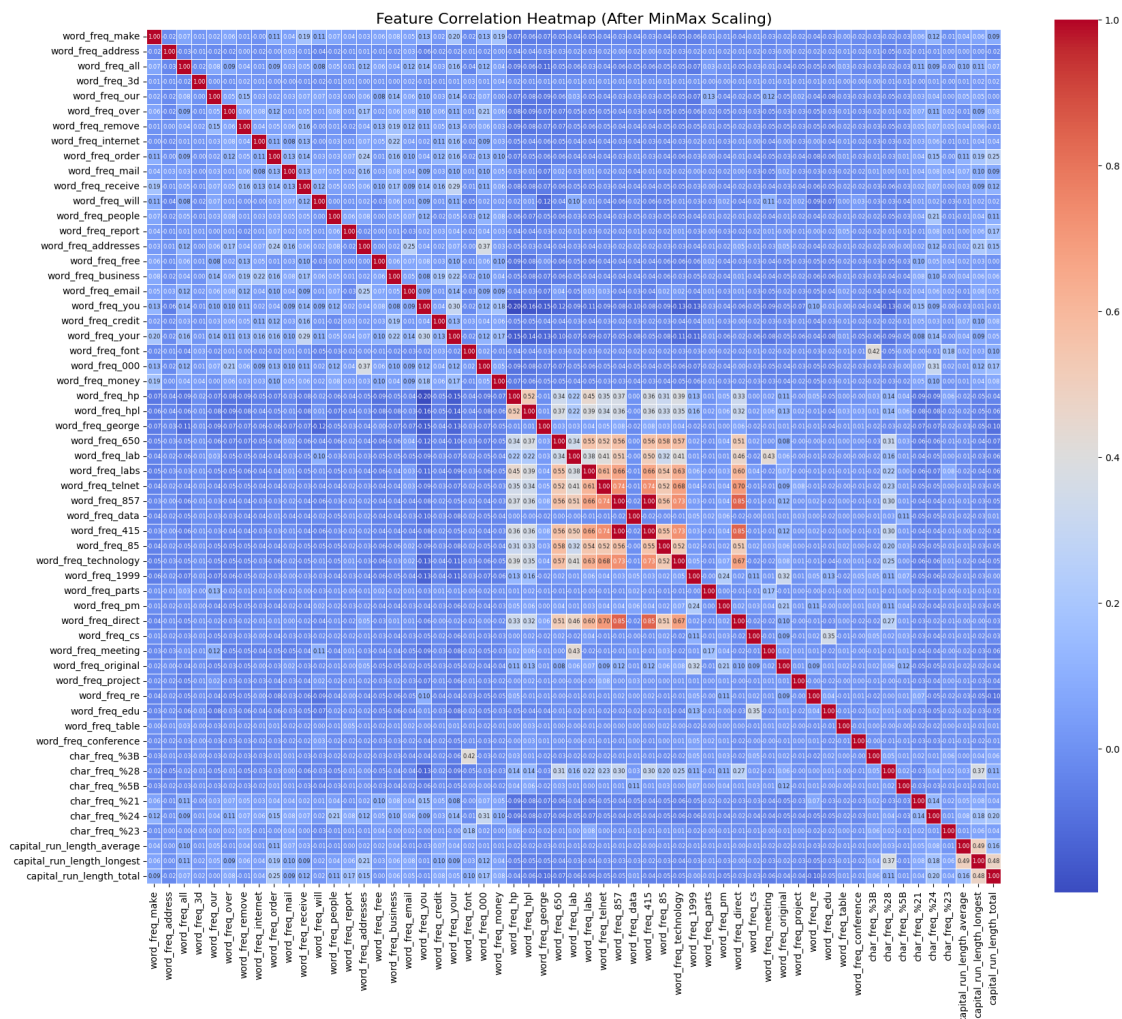
plt.figure(figsize=(18, 15)) # Bigger figure
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
```

```

square=True,
annot=True,                                     # Show values inside squares
fmt=".2f",                                       # Format to 2 decimal places
linewidths=0.5,                                 # Thin grid lines
annot_kws={"size": 6}                           # Smaller font size
)

plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.title("Feature Correlation Heatmap (After MinMax Scaling)", fontsize=16)
plt.tight_layout()
plt.show()

```



0.1.4 Model Training

```
[ ]: from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, roc_curve, auc,
    classification_report
)

# Split the data again (or reuse your earlier split)
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, stratify=y, random_state=42
)
```

```
[ ]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report, roc_curve, auc
)

# Kernels to evaluate
kernels = ['linear', 'poly', 'rbf', 'sigmoid']
results = []

def evaluate_svm(kernel):
    print(f"\n Evaluating SVM with {kernel.upper()} kernel...")

    # Create and train SVM model
    model = SVC(kernel=kernel, probability=True, random_state=42)
    model.fit(X_train, y_train)

    # Predictions
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1] # for ROC

    # Metrics
    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    print(f"Accuracy : {acc:.4f}")
    print(f"Precision: {prec:.4f}")
    print(f"Recall    : {rec:.4f}")
    print(f"F1 Score  : {f1:.4f}")
```

```

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Confusion Matrix
conf = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(5, 4))
plt.imshow(conf, cmap='Blues')
plt.title(f'{kernel.upper()} Kernel - Confusion Matrix')
plt.colorbar()
plt.xticks([0, 1], ['Not Spam', 'Spam'])
plt.yticks([0, 1], ['Not Spam', 'Spam'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
for i in range(2):
    for j in range(2):
        plt.text(j, i, conf[i, j], ha='center', va='center')
plt.tight_layout()
plt.show()

# ROC Curve
fpr, tpr, _ = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6, 5))
plt.plot(fpr, tpr, label=f'{kernel.upper()} (AUC = {roc_auc:.4f})')
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title(f'{kernel.upper()} Kernel - ROC Curve')
plt.legend()
plt.grid(True)
plt.show()

# Store results
results.append({
    "Kernel": kernel.capitalize(),
    "Accuracy": acc,
    "Precision": prec,
    "Recall": rec,
    "F1 Score": f1,
    "AUC": roc_auc
})

# Run evaluations for all kernels
for kernel in kernels:
    evaluate_svm(kernel)

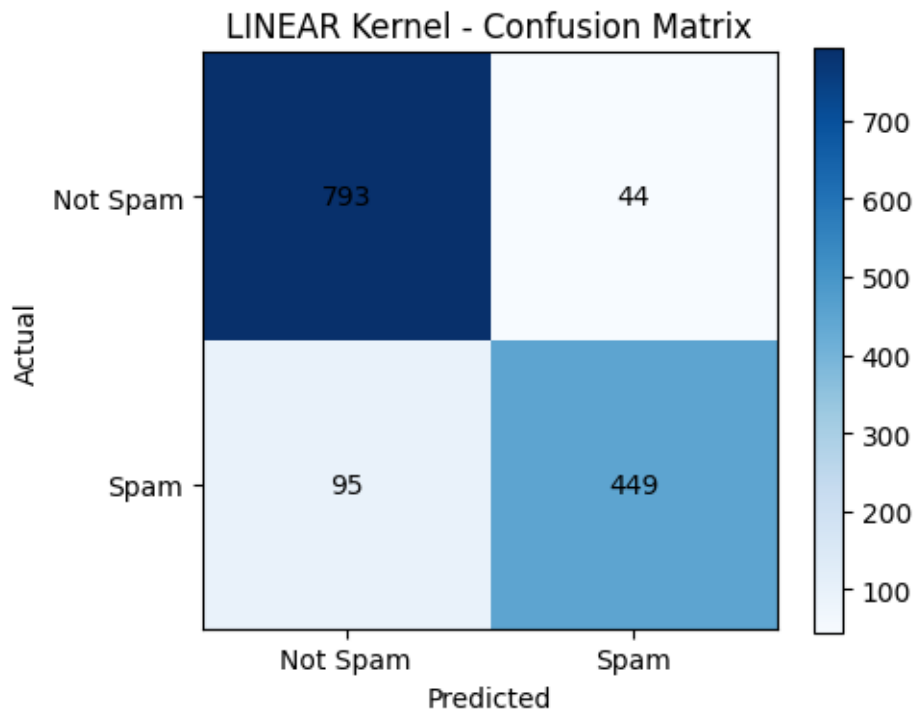
```

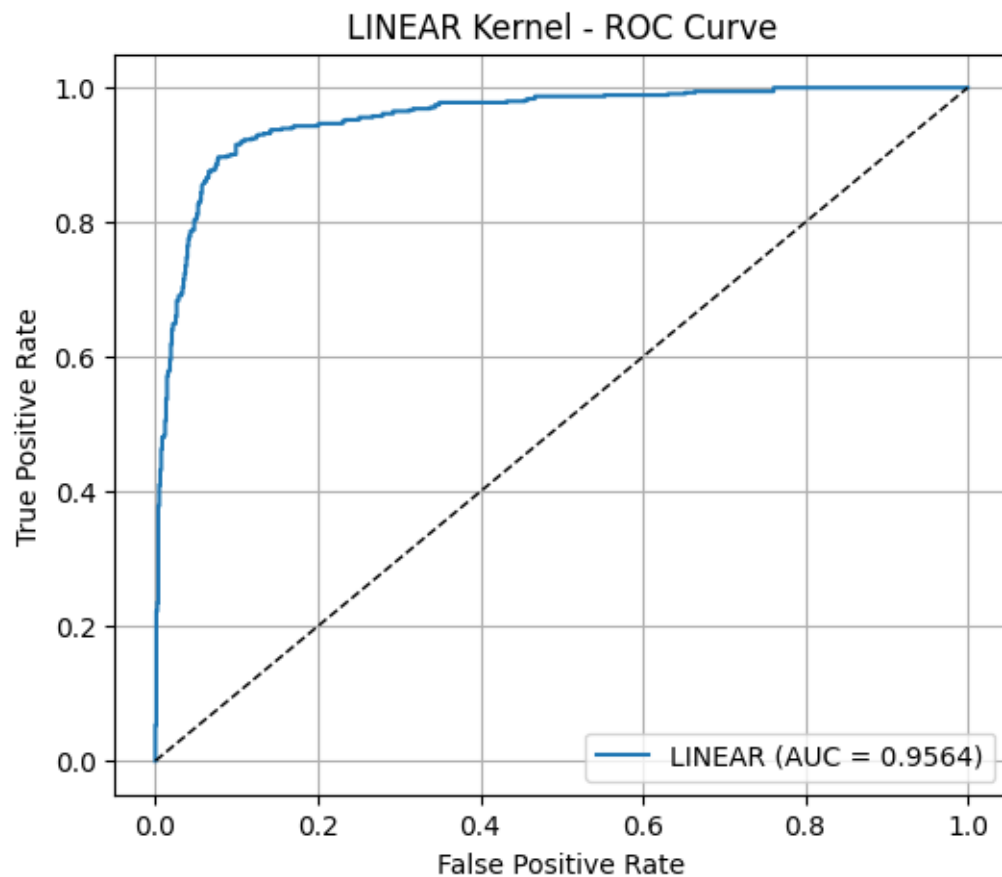
Evaluating SVM with LINEAR kernel...

Accuracy : 0.8993
Precision: 0.9108
Recall : 0.8254
F1 Score : 0.8660

Classification Report:

	precision	recall	f1-score	support
0.0	0.89	0.95	0.92	837
1.0	0.91	0.83	0.87	544
accuracy			0.90	1381
macro avg	0.90	0.89	0.89	1381
weighted avg	0.90	0.90	0.90	1381





Evaluating SVM with POLY kernel...

Accuracy : 0.8501

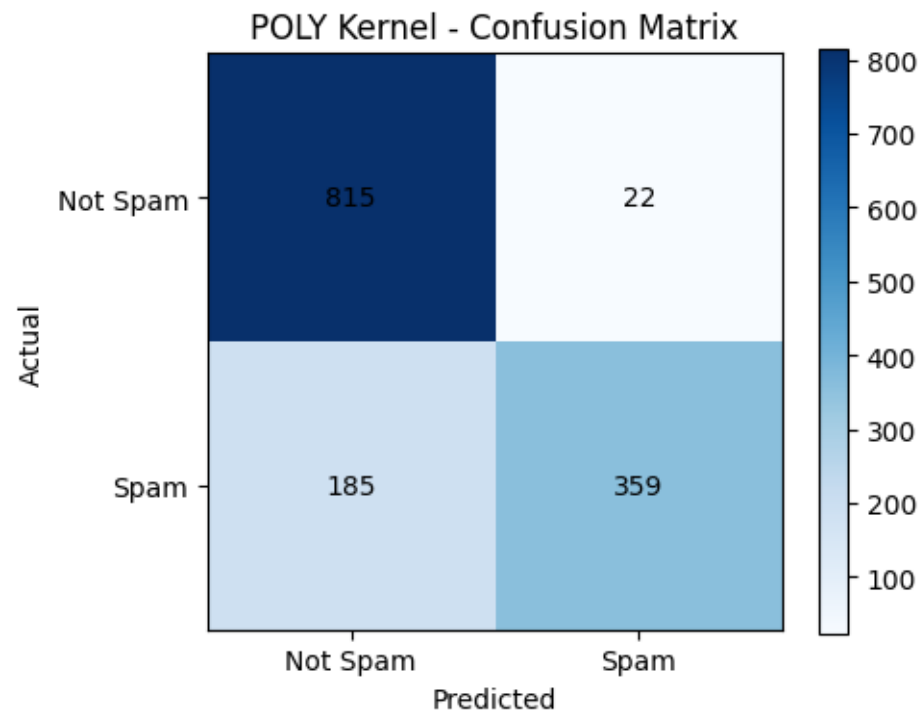
Precision: 0.9423

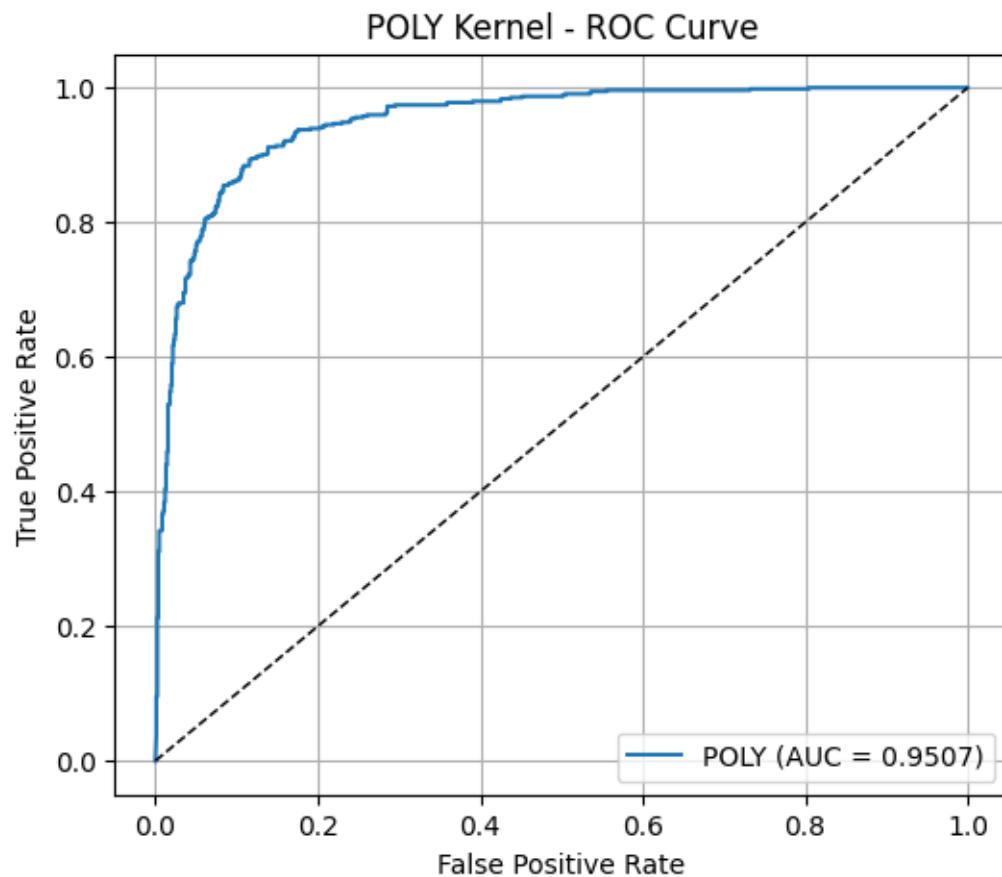
Recall : 0.6599

F1 Score : 0.7762

Classification Report:

	precision	recall	f1-score	support
0.0	0.81	0.97	0.89	837
1.0	0.94	0.66	0.78	544
accuracy			0.85	1381
macro avg	0.88	0.82	0.83	1381
weighted avg	0.87	0.85	0.84	1381





Evaluating SVM with RBF kernel...

Accuracy : 0.9261

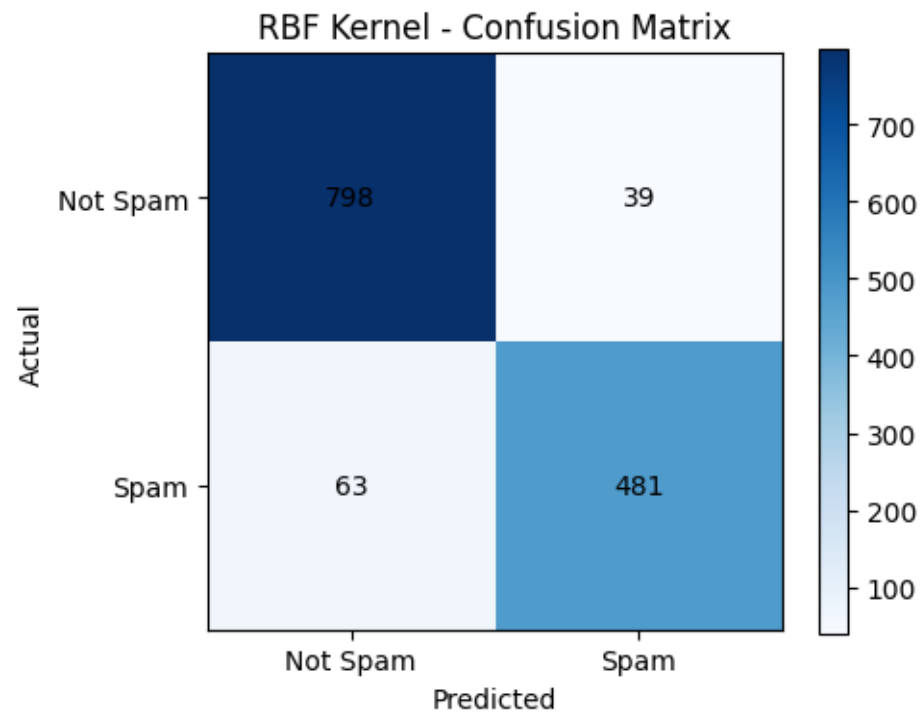
Precision: 0.9250

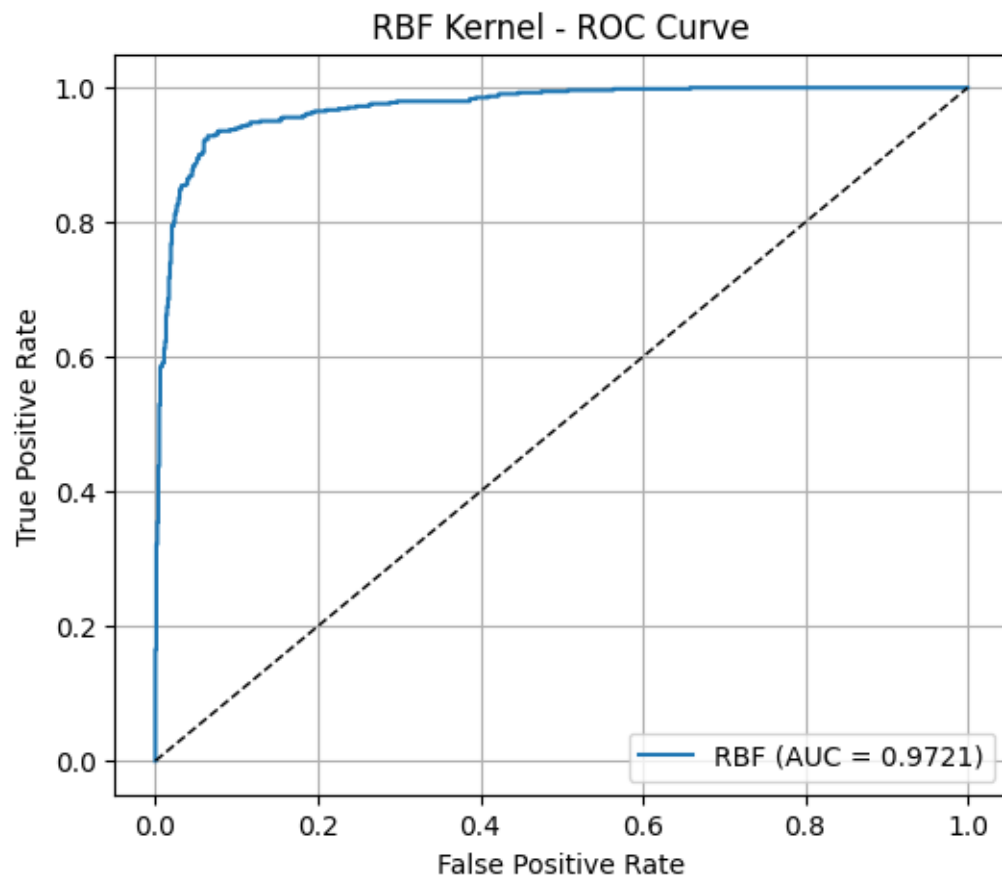
Recall : 0.8842

F1 Score : 0.9041

Classification Report:

	precision	recall	f1-score	support
0.0	0.93	0.95	0.94	837
1.0	0.93	0.88	0.90	544
accuracy			0.93	1381
macro avg	0.93	0.92	0.92	1381
weighted avg	0.93	0.93	0.93	1381





Evaluating SVM with SIGMOID kernel...

Accuracy : 0.8023

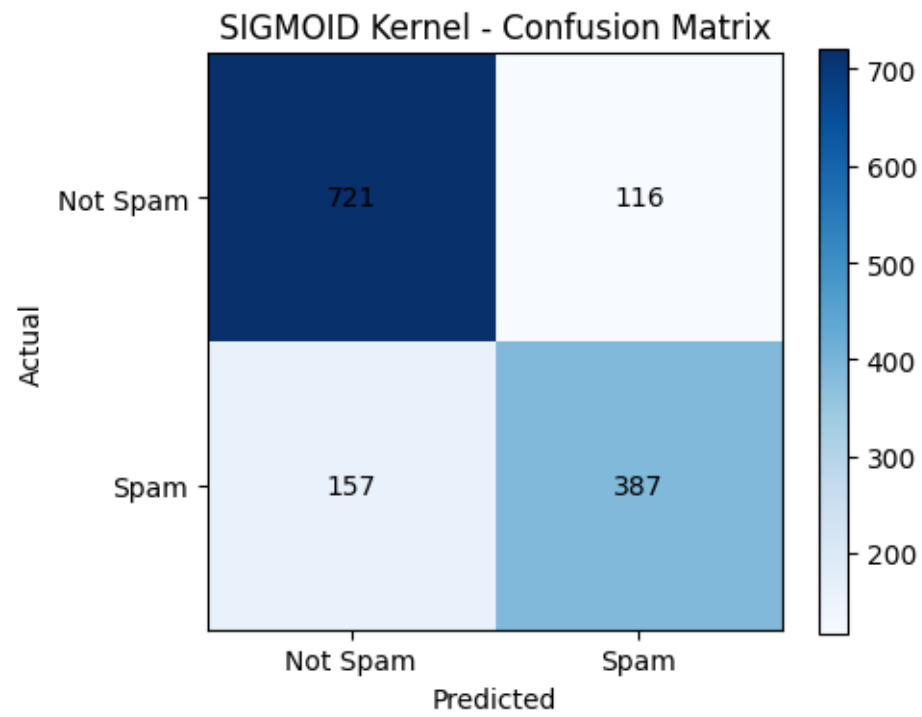
Precision: 0.7694

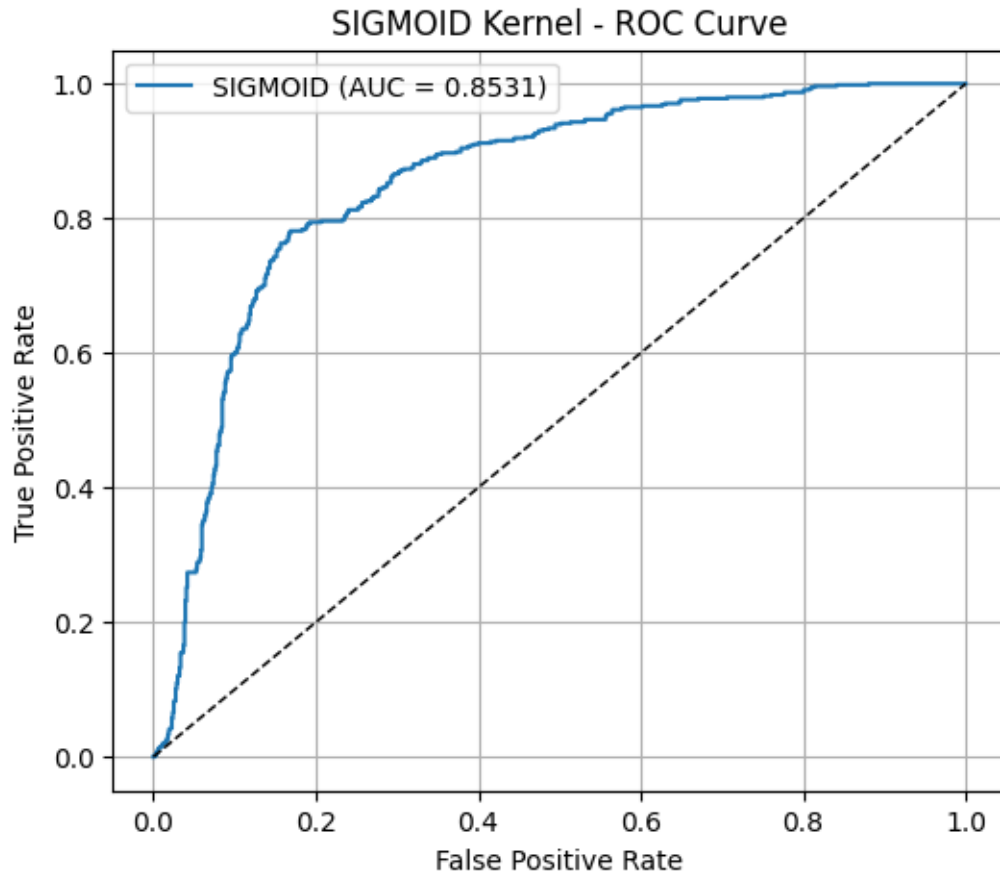
Recall : 0.7114

F1 Score : 0.7393

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.86	0.84	837
1.0	0.77	0.71	0.74	544
accuracy			0.80	1381
macro avg	0.80	0.79	0.79	1381
weighted avg	0.80	0.80	0.80	1381





```
[ ]: # Display tabular summary
print("\n SVM Kernel Comparison Summary:\n")
df_results = pd.DataFrame(results)
print(df_results.to_string(index=False))
```

SVM Kernel Comparison Summary:

Kernel	Accuracy	Precision	Recall	F1 Score	AUC
Linear	0.899348	0.910751	0.825368	0.865959	0.956425
Poly	0.850109	0.942257	0.659926	0.776216	0.950749
Rbf	0.926140	0.925000	0.884191	0.904135	0.972150
Sigmoid	0.802317	0.769384	0.711397	0.739255	0.853074

```
[ ]: svm = SVC()

param_grid = {
    'C': [0.1, 1, 10],
    'gamma': ['scale', 0.01, 0.001],
```

```

    'kernel': ['linear', 'rbf', 'poly']
}

# 5-fold CV
cv = StratifiedKfold(n_splits=5, shuffle=True, random_state=42)

# Grid Search
grid = GridSearchCV(estimator=svm, param_grid=param_grid, cv=cv,
                    scoring='accuracy', verbose=2, n_jobs=-1)
grid.fit(X_train, y_train)

# Best Parameters
print(" Best Parameters Found:")
print(grid.best_params_)

# Best Estimator
best_svm = grid.best_estimator_

# Evaluate on test set
y_pred = best_svm.predict(X_test)

print("\n Classification Report (Test Set):")
print(classification_report(y_test, y_pred))

print(" Test Accuracy:", accuracy_score(y_test, y_pred))

```

Fitting 5 folds for each of 27 candidates, totalling 135 fits

Best Parameters Found:

```
{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
```

Classification Report (Test Set):

	precision	recall	f1-score	support
0.0	0.93	0.96	0.94	837
1.0	0.93	0.88	0.91	544
accuracy			0.93	1381
macro avg	0.93	0.92	0.92	1381
weighted avg	0.93	0.93	0.93	1381

Test Accuracy: 0.9275887038377987

```
[ ]: pipeline = make_pipeline(
    MinMaxScaler(),
    SVC(kernel='rbf', C=10, gamma='scale')
)
```

```

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(pipeline, X, y, cv=cv, scoring='accuracy')

for i, acc in enumerate(scores, 1):
    print(f"Fold {i}: Accuracy = {acc:.4f}")
print(f" Mean Accuracy: {np.mean(scores):.4f} ± {np.std(scores):.4f}")

```

```

Fold 1: Accuracy = 0.9359
Fold 2: Accuracy = 0.9250
Fold 3: Accuracy = 0.9402
Fold 4: Accuracy = 0.9337
Fold 5: Accuracy = 0.9326
Mean Accuracy: 0.9335 ± 0.0050

```

```

[ ]: import time

kernels = ['linear', 'poly', 'rbf', 'sigmoid']
results = []

for kernel in kernels:
    params = {'kernel': kernel, 'C': 10.0, 'probability': True, 'random_state': 42}

    # Add kernel-specific hyperparameters
    if kernel == 'poly':
        params.update({'degree': 3, 'gamma': 'scale'})
    elif kernel in ['rbf', 'sigmoid']:
        params.update({'gamma': 'scale'})

    model = SVC(**params)

    start = time.time()
    model.fit(X_train, y_train)
    end = time.time()

    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    results.append({
        "Kernel": kernel.capitalize(),
        "Hyperparameters": str({k: v for k, v in params.items() if k != 'kernel'}),
        "Accuracy": round(acc, 4),
        "F1 Score": round(f1, 4),
        "Training Time (s)": round(end - start, 3)
    })

```

```

}))

# Create and display final table
df_table4 = pd.DataFrame(results)
print("\nTable 4: SVM Performance with Different Kernels and Parameters")
print(df_table4.to_string(index=False))

```

Table 4: SVM Performance with Different Kernels and Parameters

Kernel	Hyperparameters	Accuracy	F1 Score	Training Time (s)
Linear				{'C': 10.0, 'probability': True,
	'random_state': 42}	0.9182	0.8951	1.158
Poly	{'C': 10.0, 'probability': True, 'random_state': 42, 'degree': 3,			
	'gamma': 'scale'}	0.8979	0.8597	1.497
Rbf	{'C': 10.0, 'probability': True, 'random_state': 42,			
	'gamma': 'scale'}	0.9276	0.9058	1.628
Sigmoid	{'C': 10.0, 'probability': True, 'random_state': 42,			
	'gamma': 'scale'}	0.7864	0.7209	1.876

0.1.5 K Fold Cross Validation

```

[ ]: from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import MinMaxScaler # or StandardScaler
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.svm import SVC
import numpy as np

# RBF kernel SVM
svm_rbf = make_pipeline(MinMaxScaler(), SVC(kernel='rbf', C=10.0,
gamma='scale'))

# 5-fold stratified CV
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(svm_rbf, X, y, cv=cv, scoring='accuracy')

# Print each fold and mean
for i, acc in enumerate(scores, 1):
    print(f"Fold {i}: Accuracy = {acc:.4f}")
print(f"\n Mean Accuracy: {np.mean(scores):.4f} ± {np.std(scores):.4f}")

```

Fold 1: Accuracy = 0.9359
 Fold 2: Accuracy = 0.9250
 Fold 3: Accuracy = 0.9402
 Fold 4: Accuracy = 0.9337
 Fold 5: Accuracy = 0.9326

Mean Accuracy: 0.9335 ± 0.0050

7 Comparison Tables

Table 1: Performance Comparison of Naïve Bayes Variants

Model	Accuracy	Precision	Recall	F1 Score
BernoulliNB	0.8899	0.8843	0.8290	0.8558
MultinomialNB	0.8950	0.9424	0.7813	0.8543
GaussianNB	0.8197	0.7001	0.9485	0.8056

Table 2: KNN Performance for Different k Values

k	Accuracy	Precision	Recall	F1 Score
1	0.8899	0.8551	0.8676	0.8613
3	0.8892	0.8710	0.8438	0.8571
5	0.8993	0.8828	0.8585	0.8705
7	0.8950	0.8815	0.8474	0.8641

Best Hyperparameters:

k value: 1

Table 3: KNN Comparison: KDTree vs BallTree

Metric	KDTree	BallTree
Accuracy	0.8899	0.8899
Precision	0.8551	0.8551
Recall	0.8676	0.8676
F1 Score	0.8613	0.8613
Training Time (s)	0.4470	0.4058

Table 4: SVM Performance with Different Kernels and Parameters

Kernel	Hyperparameters	Accuracy	F1 Score	Training Time (s)
Linear	C: 10.0	0.9182	0.8951	1.158
Polynomial	C: 10.0, degree: 3, gamma: 'scale'	0.8979	0.8597	1.497
RBF	C: 10.0, gamma: 'scale'	0.9276	0.9058	1.628
Sigmoid	C: 10.0, gamma: 'scale'	0.7864	0.7209	1.876

Best Hyperparameters:

Kernel: Linear
C: 10.0

Table 5: Cross-Validation Scores for Each Model (K = 5)

Fold	Naïve Bayes Accuracy	KNN Accuracy (k=1)	SVM Accuracy (RBF)
Fold 1	0.8719	0.9034	0.9359
Fold 2	0.8935	0.9076	0.9250
Fold 3	0.8891	0.9152	0.9402
Fold 4	0.8913	0.9000	0.9337
Fold 5	0.8859	0.8935	0.9326
Average	0.8863	0.9039	0.9335

8 Observation:

- KNN (k=1) consistently achieved higher accuracy across all 5 folds compared to Multinomial Naïve Bayes, with an average accuracy of 90.39 percentage vs 88.63 percentage.
- Naïve Bayes showed slightly more stable performance across folds, whereas KNN had slightly higher variance but better peak performance (e.g., Fold 3 with 91.52 percentage).
- KNN's superior accuracy suggests that instance-based learning works better for this dataset, possibly due to well-separated class boundaries that KNN can capture using distance metrics.
- RBF kernel gives the highest accuracy (0.9276) and best F1 (0.9058), slightly slower than Linear but much better performance.

9 Conclusion:

- KNN (k=1) outperformed Multinomial Naïve Bayes in terms of average accuracy in 5-fold cross-validation, making it the better choice for this dataset.
- However, Naïve Bayes remains a strong baseline due to its simplicity, speed, and competitive accuracy, especially for high-dimensional or text data.
- SVM with the RBF kernel achieved the highest overall accuracy and F1 score, showing superior generalization across folds, making it the most reliable model among all tested.

GitHub Link